C2FNet: Cross-Probabilistic Weak Supervision Learning for High-Resolution Land Cover Enhancement

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***Abstract*—Autonomous large-scale high-resolution land cover (HRLC) mapping remains a major challenge in remote sensing due to the scarcity of reliable training data and resolution mismatches between available labels and input of massive of emerging imagery. Existing global land cover products often suffer from coarse spatial resolution and label noise, limiting their utility for fine-scale urban analysis and environmental monitoring. This article presents C2FNet, a novel Coarse-to-Fine Network designed to generate HRLC maps from noisy, coarse-resolution labels using a weak supervision strategy of the cross-probability. The C2FNet consists of three key modules: 1) edge resolution refinement backbones (ERRBs), which preserve spatial detail via multiscale feature extraction through parallel convolutional branches; 2) unsupervised dynamic shuffle and diagonal annotation (UDSDA), which enhances training reliability by identifying confident regions through spatial-consistency analysis and confidence estimation; and 3) a contrasting self-supervised loss (C2F-Loss) that integrates cross-entropy and cosine similarity terms to mitigate supervision noise and resolution gaps. Evaluations of three benchmark datasets that encompass diverse urban and rural landscapes show that C2FNet achieves state-of-the-art (SoA) performance, with 80.01% overall accuracy (OA) and a Cohen's kappa score of 0.7567, outperforming SoA models with weak supervision. The dataset and code are available at http://drive.google.com/file/d/1X_Fz7LQIeix3rV3K29FBfKiU1WMdROe-/view**

***Index Terms*—Coarse-to-fine networks (C2FNets), cross-resolution learning, deep neural networks, Earth observation, land cover mapping, probabilistic supervision, remote sensing, weakly supervised learning (WSL).**

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I. Introduction

LAND cover mapping represents one of the most critical applications in Earth observation, providing essential information for environmental monitoring [1], urban planning, climate change assessment, and sustainable development initiatives [2], [3]. The increasing availability of high-resolution (HR) satellite imagery from platforms such as WorldView, QuickBird, and emerging constellations presents unprecedented opportunities for detailed analysis of the Earth's surface [4], [5]. However, the fundamental challenge of acquiring corresponding high-quality training labels at fine spatial resolutions continues to impede the full utilization of these rich data sources [6], [7]. Urban planning, climate change assessment, and sustainable development initiatives increasingly benefit from geospatial data analytics [8], [9]. The growing availability of HR satellite imagery from platforms such as WorldView, QuickBird, and emerging constellations presents unprecedented opportunities for detailed analysis of the Earth's surface [10], [11]. However, the fundamental challenge of acquiring high-quality training labels at fine spatial resolutions continues to impede the full utilization of these rich data sources [12], [13]. Current global land cover products, including those from the European Space Agency Climate Change Initiative (ESA-CCI), the Global Land Cover Facility (GLCF), and commercial providers, typically operate at 10–30-m spatial resolution [14], [15]. While these products provide valuable global coverage, they contain substantial label noise and lack the spatial detail necessary for fine-scale

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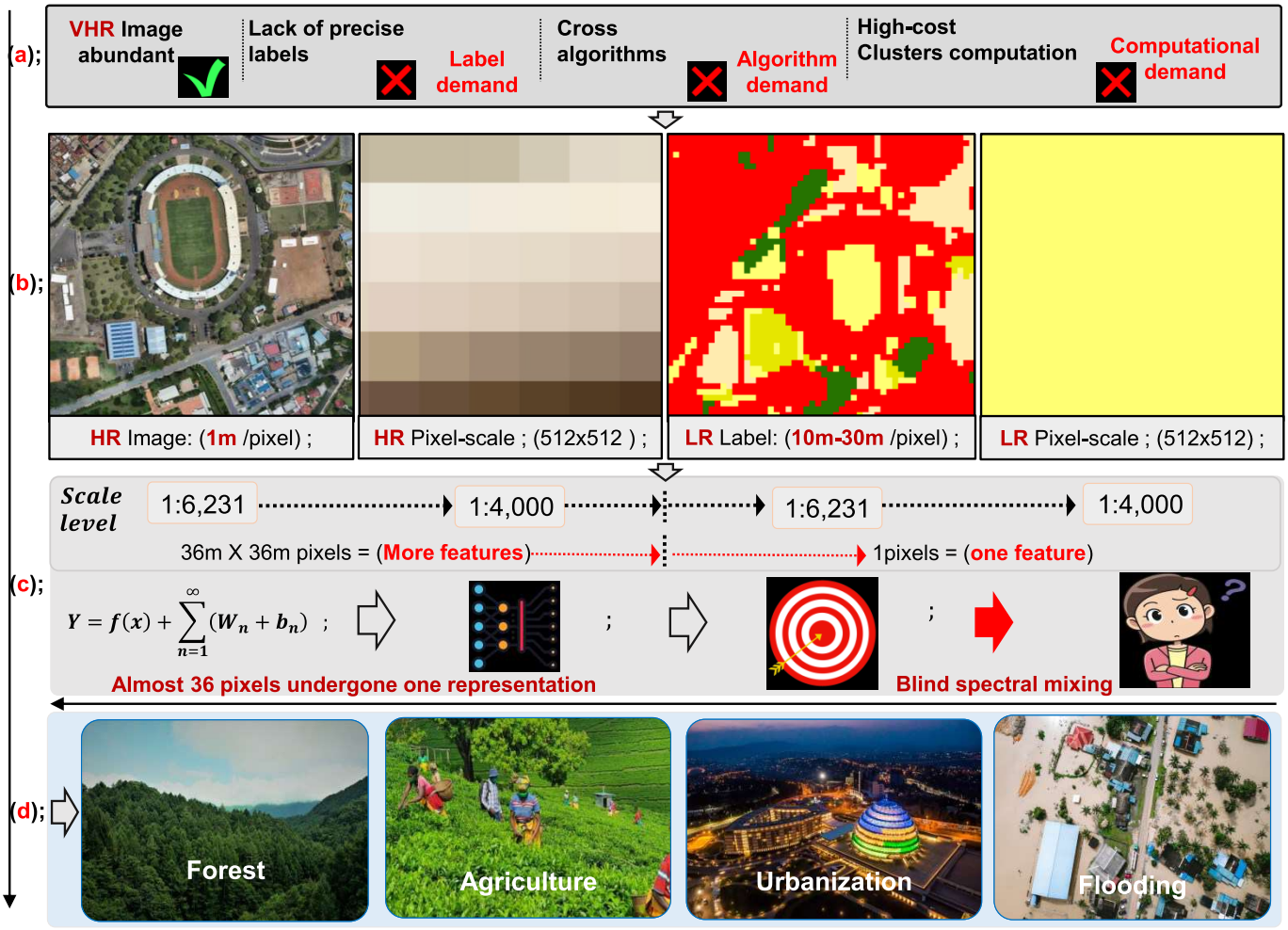


Fig. 1. Illustration of the necessity for local-scale, HRLC products for fine-scale urban–rural analytics. (a) Top portion highlights challenges such as label inconsistencies, algorithmic disparities, and high computational complexity due to spatial variations among products at the same scale. (b) Middle section demonstrates the loss of critical detail due to blind hyperspectral mixing, resulting in overly generalized features. (c) Bottom portion presents the objectives of this study, aimed at scalable real-world applications. This graphical abstract emphasizes implicit knowledge acquisition and addresses the spatial resolution mismatch challenge: using HR remote sensing imagery together with pseudo-low-resolution (PLR) historical labels (10–30 m) to generate HR land cover maps.

urban analysis, precision agriculture, and local environmental management [16], [17].

Fig. 1 illustrates the coarse resolution of existing products creates a significant bottleneck for applications requiring meter or submeter level detail, particularly in heterogeneous urban environments where land cover patterns change rapidly over short distances [18], [19]. The fundamental challenge lies in what we termed the "resolution mismatch problem" the disparity between available HR imagery (often submeter to few meters) and existing training labels (typically 10–30-m resolution) [20], [21]. This disparity creates a semantic gap where fine spatial details visible in HR images cannot be properly supervised by coarse labels [22], [23], leading to suboptimal model performance and loss of critical spatial information, as illustrated in Fig. 1. Traditional approaches to address this problem include super-resolution techniques, which attempt to enhance coarse labels, and multiscale fusion methods, which combine information from different resolution sources [24], [25], [26]. However, these approaches often make unrealistic assumptions about the relationship between different resolutions or require additional HR labeled data that are expensive to acquire [27], [28]. Traditional supervised learning approaches require pixel-perfect labels, which are expensive and time-consuming to acquire at scale [29]. Manual annotation of HR imagery for entire cities or regions is prohibitively costly, requiring specialized expertise and enormous human effort. For instance, creating pixel-level annotations for a single 1-km^2 urban area at 1-m resolution requires labeling over 1 million pixels across multiple land cover classes. Automated labeling methods, while faster, often introduce systematic errors and fail to capture the complexity of real-world land cover patterns [30]. Weak supervision learning offers a promising alternative by leveraging imperfect, noisy, or limited supervision signals instead of requiring perfect pixel-level annotations [15], [18]. This paradigm has gained significant attention in computer vision and natural language processing, where it has been successfully applied to tasks such as image classification, object detection, and text analysis

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[48], [49]. In the context of remote sensing, weak supervision can potentially bridge the gap between available coarse-resolution labels and the need for fine-resolution predictions.

Recent advances in deep learning have demonstrated the remarkable effectiveness of convolutional neural networks (CNNs) for remote sensing applications [15]. Classical architectures in segmentation networks such as UNet [5], DeepLabV3+ [17], and feature pyramid networks (FPNs) [20] have achieved state-of-the-art (SoA) performance across various remote sensing tasks. However, most existing methods assume the availability of high-quality training labels at the target resolution. When applied to scenarios with significant resolution mismatches, these methods often produce blurry or inconsistent results due to inadequate supervision at fine scales. The emergence of Vision Transformers (ViTs) [21] has introduced new possibilities for handling multiscale information through self-attention mechanisms [61]. However, ViTs require enormous amounts of training data and computational resources [47], [48], making them less suitable for scenarios with limited labeled data typical in remote sensing applications [49]. Recent advances in weakly supervised learning (WSL) [31], [50] and self-training methods [58] have shown promise for addressing annotation scarcity, but these approaches often struggle with the specific challenges of cross-resolution supervision in Earth observation data [51].

This article addresses the aforementioned limitations by introducing coarse-to-fine network (C2FNet), a novel framework that explicitly handles resolution mismatches through cross-probabilistic weak supervision. Our approach is motivated by the observation that while coarse labels may lack spatial detail, they still contain valuable semantic information that can guide the learning process when combined with appropriate architectural innovations [53] and loss functions [54]. The key insight behind C2FNet is that the resolution mismatch problem can be addressed by preserving spatial information throughout the network architecture [55], automatically identifying reliable training regions within noisy coarse labels [57], and designing loss functions that can handle the inherent uncertainty in cross-resolution supervision [56]. This leads to a framework that can effectively learn from readily available coarse-resolution labels while producing high-quality fine-resolution predictions [59], [60].

The main contributions of this article are summarized as follows.

1) We propose a novel weakly supervised deep learning framework, termed C2FNet, which refines coarse-resolution annotations into fine-resolution land cover predictions.
2) We introduce the edge resolution refinement backbones (ERRBs), a specialized architectural component designed to preserve spatial detail while extracting rich multiscale features through parallel convolutional branches with varied receptive fields.
3) We develop an unsupervised region selection mechanism, unsupervised dynamic shuffle and diagonal annotation (UDSDA), which identifies reliable training areas from noisy supervision using confidence estimation and spatial pattern reasoning.
4) We design a hybrid self-supervised loss function, contrasting self-supervised loss (C2F-Loss), that combines cross-entropy with cosine similarity. This loss promotes stable gradient propagation and enhances semantic consistency under weak supervision settings.
5) We conduct extensive experiments on three HR geospatial datasets representing urban and rural regions. The results demonstrate that C2FNet consistently outperforms SoA methods, achieving both high segmentation accuracy and efficient computational performance.

II. Related Work

A. Deep Learning for Land Cover Mapping

Deep learning has fundamentally transformed the landscape of remote sensing and land cover mapping over the past decade [15]. The introduction of CNNs has enabled automatic feature extraction and end-to-end learning [47], [48], replacing traditional handcrafted feature approaches that dominated the field for decades [62]. UNet [5], originally designed for biomedical image segmentation, has become one of the most widely adopted architectures for land cover mapping [6]. Its symmetric encoder–decoder structure with skip connections enables effective multiscale feature fusion and has been successfully applied to various remote sensing tasks, including building extraction, road detection, and forest monitoring [63], [64]. However, UNet's reliance on aggressive downsampling in the encoder can lead to loss of fine spatial details, particularly problematic for HR land cover (HRLC) mapping [65]. DeepLabV3+ [17] introduced atrous (dilated) convolution and atrous spatial pyramid pooling (ASPP) to capture multiscale contextual information without reducing spatial resolution [16]. This approach has shown particular effectiveness for scenes with objects at multiple scales, making it well-suited for urban land cover mapping where features range from individual buildings to large forest patches [66]. The encoder–decoder structure of DeepLabV3+ [17] with its lightweight decoder has achieved SoA performance on several benchmark datasets [67], [68]. SegNet [18] represents another influential architecture that uses pooling indices to transfer spatial information from encoder to decoder, reducing the number of parameters while maintaining segmentation accuracy. This efficiency makes SegNet attractive for large-scale applications where computational resources are limited [49]. More recent architectures have focused on improving multiscale feature extraction and boundary delineation [69]. FPNs [20] construct feature pyramids through top-down pathways and lateral connections, enabling detection and segmentation of objects at multiple scales. LinkNet [19] provides an efficient alternative with fewer parameters while maintaining competitive performance. The recent introduction of ViTs [21] has opened new possibilities for remote sensing applications [61]. ViTs [21] treat images as sequences of patches and apply self-attention mechanisms to capture long-range dependencies. While promising, ViTs typically require enormous amounts of training data and computational resources, limiting their applicability in remote sensing scenarios where labeled data are scarce. Despite these architectural advances, most existing

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methods assume the availability of high-quality training labels at the target resolution. When applied to cross-resolution scenarios, these methods often struggle to maintain fine spatial detail while leveraging coarse supervision effectively.

B. WSL in Remote Sensing

WSL has become an essential strategy in remote sensing, addressing the prohibitive cost and complexity of acquiring dense, pixel-level annotations at large scales [8], [9], [12], [14]. Common WSL forms include image-level labels [22], [24], [25], scribbles [55], bounding boxes, pseudo-labels (PLs), and noisy annotations. While each approach lowers the annotation burden, they pose unique challenges for dense prediction tasks such as land cover mapping.

Image-level supervision, among the weakest forms, uses global labels and leverages techniques like class activation mapping (CAM) to localize objects [22]. However, these often fail to capture the complete object extent due to their focus on the most discriminative regions. In remote sensing, such methods have been applied to object detection [24] and classification [25], but they struggle with the semantic gap between image-level cues and pixel-level output. Scribble-based supervision [55] offers slightly better localization but still requires manual interaction.

Learning under label noise, a common issue in weak supervision, has been tackled using co-teaching [23], meta-learning, and consistency regularization [31]. Programmatic labeling approaches like data programming [14] also aim to automate noisy label generation using rules and heuristics. However, most of these techniques emphasize object detection or classification, not full-scene segmentation. Recent research has begun exploring WSL specifically for dense land cover prediction using coarse-to-fine label supervision, where fine-resolution images are supervised by coarse-resolution labels. Representative works include L2HNet [8], which learns to generate HRLC from low-resolution inputs, and Paraformer [36], which integrates HR priors with low-resolution labels using deformable attention. Similarly, the SinoLC-1 project [8] demonstrates national-scale mapping with deep learning and open-access coarse labels. Despite their promise, these approaches are still limited by reliance on model scale or hand-crafted refinement rules. Our work advances this direction by introducing C2FNet, a framework tailored to cross-resolution weak supervision. Unlike traditional WSL that treats label noise as random, we exploit the structured nature of resolution mismatch. Therefore, among core innovations includes an ERRB for spatial detail preservation and UDSDA module to isolate clean supervision regions, which makes the framework uniquely positioned to address the practical reality of abundant low-resolution land cover products and scarce HR annotations.

C. Multiresolution Learning and Domain Adaptation

The challenge of handling multiple resolutions in remote sensing has been addressed through various approaches, including super-resolution, multiscale fusion, and domain adaptation techniques [26], [27]. Super-resolution methods attempt to enhance the spatial resolution of coarse imagery or labels through interpolation, reconstruction, or learning-based approaches. While effective for enhancing image quality, these methods often introduce artifacts and may not preserve the statistical properties of land cover distributions. Multiscale fusion techniques combine information from sensors or datasets with different spatial resolutions [13]. These approaches typically require careful alignment and calibration of multisource data, and their effectiveness depends on the availability of complementary information across different resolutions. Domain adaptation methods have been explored to transfer knowledge from coarse to fine resolutions [29]. These approaches treat different resolutions as different domains and employ adversarial training or other domain adaptation techniques to bridge the resolution gap. However, they often require additional HR labeled data for effective adaptation.

Recent studies have explored generative adversarial networks (GANs) for cross-resolution learning, where generators synthesize fine-resolution outputs from coarse inputs and discriminators enforce spatial realism [28]. While promising, such methods often suffer from training instability and the generation of unrealistic fine-scale patterns, particularly in heterogeneous landscapes [58]. Despite these advances, key challenges remain. Most existing approaches do not directly address the core problem of producing fine-resolution outputs from coarse and noisy labels [23]. Although weak supervision has been applied in remote sensing, it often lacks robustness against spatial and semantic inconsistencies inherent in real-world datasets [22], [25]. Furthermore, many models demand substantial computational resources [30], limiting their deployment at scale. Existing architectures also struggle to balance multiscale context integration with the preservation of fine spatial details [55], and few of them demonstrate effectiveness in operational scenarios [29]. C2FNet bridges these gaps through a unified framework that integrates architectural innovations [53], adaptive sample selection [57], and a resolution-aware loss function [56] specifically tailored for weakly supervised cross-resolution land cover mapping in remote sensing applications.

III. Methodology

A. Problem Formulation

Let $\mathcal{X} = \{X_i\}_{i=1}^{N}$ denote a collection of HR remote sensing images, where each image $X_i \in \mathbb{R}^{H\times W\times C}$ has spatial dimensions $H \times W$ and C spectral channels at fine spatial resolution r_h (typically 0.5–2 m). Let $\mathcal{Y} = \{Y_i\}_{i=1}^{N}$ represent the corresponding coarse-resolution land cover labels, where each label map $Y_i \in \{1, 2, \ldots, K\}^{H'\times W'}$ contains K land cover classes at coarse spatial resolution r_c (typically 10–30 m), where $r_h \ll r_c$.

As Fig. 1 illustrates, the fundamental challenge is the significant resolution mismatch where $(r_c/r_h) \gg 1$, meaning that each coarse label pixel corresponds to multiple HR image pixels. This creates an inherent ambiguity in supervision where fine-scale spatial patterns visible in X_i cannot be directly supervised by the spatially averaged information in Y_i. Our objective is to learn a mapping function $f_\theta : \mathbb{R}^{H\times W\times C} \rightarrow \mathbb{R}^{H\times W\times K}$ parameterized by θ that produces HRLC probability maps

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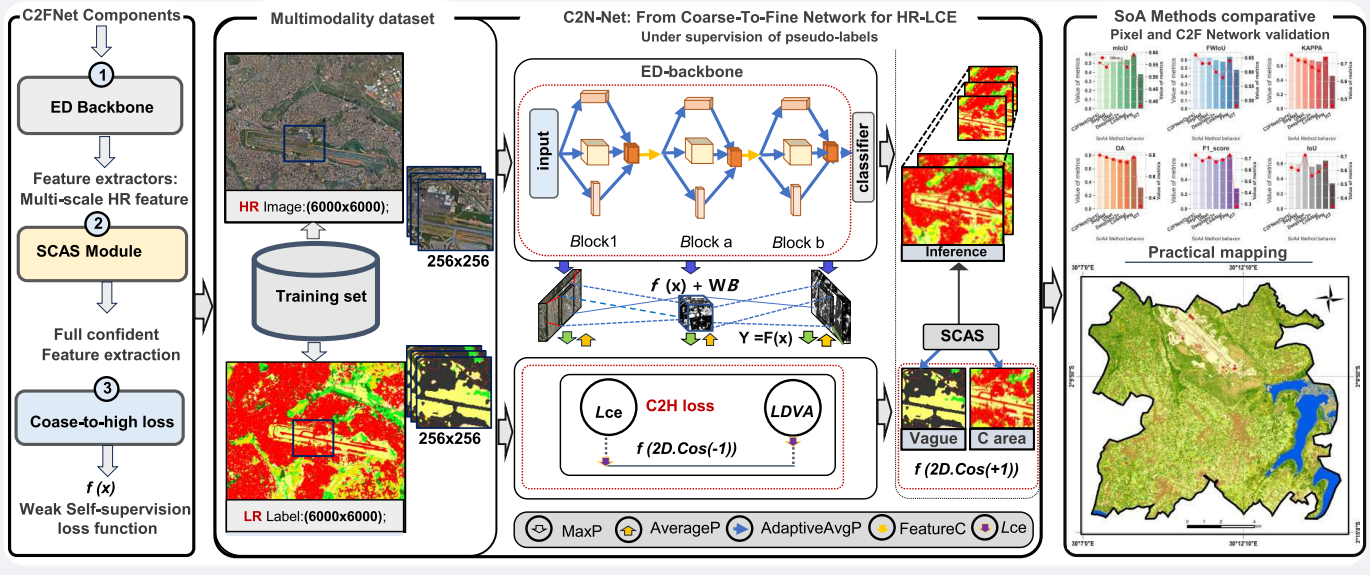


Fig. 2. Overview of the C2FNet architecture designed for HRLC estimation. The framework comprises five core components: 1) identification of architectural bottlenecks motivating the C2FNet design; 2) a data engineering pipeline that generates over 12000 weakly supervised training samples from multimodal inputs; 3) ERRB with the proposed coarse-to-high (C2H) loss function, enabling fine-grained predictions from coarse PLs; 4) the self-calibrating accuracy selection (SCAS) module and confidence accuracy module for improved feature refinement and confidence-aware mapping; and 5) comparative visualizations and quantitative evaluations demonstrating the effectiveness of probabilistic weak supervision learning (PWSL) strategies against SoA baselines.

$\hat{Z}_i = f_\theta(X_i)$ at the same spatial resolution as the input imagery, as shwon in Fig. 2. The key challenge lies in effectively utilizing the coarse supervision $\mathcal{Y}$ to train f_θ despite the resolution mismatch.

B. Network Overview

The proposed C2FNet network, as illustrated in Fig. 2, addresses the resolution mismatch problem through three components designed to work in concert. The ERRBs preserve spatial information while extracting features at multiple scales through parallel convolutional paths. The UDSDA automatically identifies reliable supervision regions within coarse labels through confidence estimation. The C2F-Loss handles supervision uncertainty through a combination of classification and feature alignment objectives. Each component addresses a specific aspect of the resolution mismatch problem, and their integration enables effective learning from coarse supervision while producing fine-resolution outputs.

C. Edge Resolution Refinement Backbones (ERRBs)

Traditional CNN architectures for semantic segmentation employ aggressive downsampling in encoder networks to build hierarchical feature representations. While this approach is effective when training and inference resolutions match, it becomes problematic in cross-resolution scenarios where fine spatial details must be preserved. The ERRB module addresses this limitation by maintaining the spatial resolution throughout the network while capturing multiscale contextual information through parallel convolutional branches with different receptive fields. Fig. 3 demonstrates the ERRB module, which consists of M parallel convolutional branches, each designed

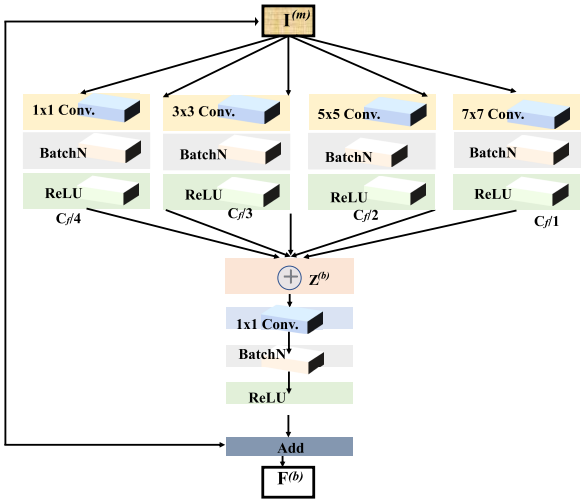


Fig. 3. ERRB module architecture demonstrating multiscale feature extraction using convolutional layers with kernel sizes of 1×1, 3×3, 5×5, and 7×7. All kernels use odd dimensions to maintain spatial alignment and preserve center pixel positioning. The feature maps are processed through parallel branches with different receptive fields, followed by attention-based fusion. The convolution kernels are optimized to prevent excessive feature resolution reduction from over-downsampling while capturing multiscale contextual information essential for accurate land cover classification.

to capture features at different scales. The multiscale parallel processing is achieved through

$$F_k^{(l)} = \text{Conv}_{k\times k}\big(\text{BN}\big(\text{ReLU}\big(F^{(l-1)}\big)\big)\big) \tag{1}$$

where $F_k^{(l)}$ represents the feature map produced by the $k \times k$ convolutional layer at level l, $k \in \{1, 3, 5, 7\}$ represents different kernel sizes for capturing multiscale patterns, BN denotes the batch normalization, and ReLU is the rectified linear activation function.

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To prevent excessive parameter growth while maintaining representational capacity, we employed depthwise separable convolutions for larger kernel sizes

$$F_k^{(l)} = \mathrm{PConv}_{1\times1}\left(\mathrm{DConv}_{k\times k}\left(\mathrm{BN}\left(\mathrm{ReLU}\left(F^{(l-1)}\right)\right)\right)\right) \quad (2)$$

where DConv represents the depthwise convolution and PConv represents the pointwise convolution.

The multiscale features are combined through a learned attention mechanism that adaptively weights the contribution of different scales through adaptive feature fusion

$$\alpha_k = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times1}\left(\mathrm{GAP}\left(F_k^{(l)}\right)\right)\right) \quad (3)$$

$$F_{\mathrm{fused}}^{(l)} = \sum_k \alpha_k \odot F_k^{(l)} \quad (4)$$

where GAP denotes the global average pooling, α_k represents the attention weights for scale k, and $\odot$ denotes the element-wise multiplication.

To preserve fine spatial details crucial for HRLC mapping, we incorporated an edge enhancement module that explicitly models the boundary information

$$E = \mathrm{Conv}_{3\times3}\left(\left|\nabla_x F_{\mathrm{fused}}^{(l)}\right| + \left|\nabla_y F_{\mathrm{fused}}^{(l)}\right|\right) \quad (5)$$

$$F_{\mathrm{revised}}^{(l)} = F_{\mathrm{fused}}^{(l)} + \gamma \cdot E \quad (6)$$

where ∇_x and ∇_y represent the spatial gradients in horizontal and vertical directions, respectively, and γ is a learnable parameter controlling the strength of edge enhancement. The detailed architecture of the ERRB module is shown in Fig. 3.

D. Unsupervised Dynamic Shuffle and Diagonal Annotation

Fig. 4 demonstrates the UDSDA module, which addressed the challenge of label noise and spatial uncertainty inherent in coarse-resolution supervision. Rather than treating all coarse labels as equally reliable, UDSDA automatically identifies regions where coarse labels provide trustworthy supervision for fine-resolution learning. The confidence estimation process begins by analyzing the consistency of predictions across multiple scales and spatial neighborhoods. For each spatial location (i, j) in the feature map, we computed a confidence score based on prediction stability

$$s_{ij}^{(k)} = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times1}\left(F_k^{(l)}(i,j)\right)\right) \quad (7)$$

$$C_{ij} = 1 - \frac{1}{M-1}\sum_{k=1}^{M}\sum_{k'=k+1}^{M}\mathrm{KL}\left(s_{ij}^{(k)} \| s_{ij}^{(k')}\right) \quad (8)$$

where $s_{ij}^{(k)}$ represents the softmax probability distribution at location (i, j) for scale k, KL denotes the Kullback–Leibler divergence, and C_{ij} measures the consistency of predictions across different scales.

In addition to multiscale consistency, we analyzed local spatial patterns to identify homogeneous regions where coarse labels are more likely to be reliable through spatial pattern analysis

$$H_{ij} = \exp\left(-\frac{1}{|\mathcal{N}_{ij}|}\sum_{(p,q)\in\mathcal{N}_{ij}}\left\|F_{ij} - F_{pq}\right\|_2^2\right) \quad (9)$$

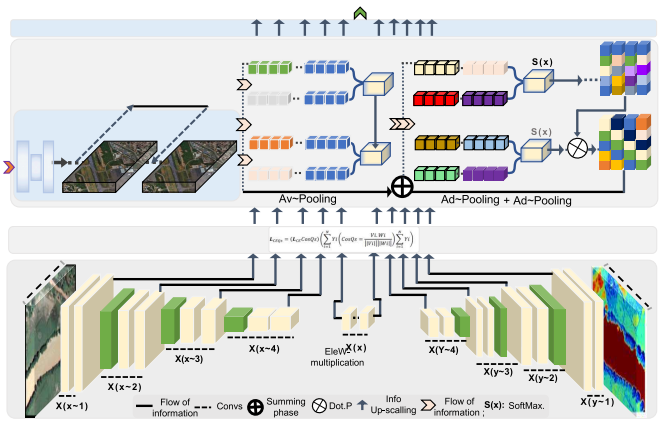


Fig. 4. UDSDA process showing the confidence estimation and spatial pattern analysis for reliable sample selection.

where $\mathcal{N}_{ij}$ represents the spatial neighborhood around location (i, j) and H_{ij} measures the local homogeneity in the feature space.

The final confidence mask is generated through adaptive thresholding that accounts for both prediction consistency and spatial homogeneity via dynamic confidence thresholding

$$\tilde{C}_{ij} = \beta C_{ij} + (1-\beta)H_{ij} \quad (10)$$

$$M_{ij} = \begin{cases} 1, & \text{if } \tilde{C}_{ij} > \tau_t \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

where β balances the contributions of consistency and homogeneity and τ_t is a time-varying threshold that starts high and gradually decreases during training to progressively include more challenging samples

$$\tau_t = \tau_0 \cdot \exp(-\lambda t) \quad (12)$$

where τ_0 is the initial threshold, λ controls the decay rate, and t represents the training iteration. The UDSDA process is illustrated in Fig. 4.

E. Contrasting Self-Supervised Loss

The C2F-Loss function is designed to handle the inherent uncertainty in cross-resolution supervision while maintaining training stability and encouraging meaningful feature representations. The traditional cross-entropy loss is modified to only include reliable regions identified by UDSDA and weighted by confidence scores through confidence-weighted cross-entropy loss

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{|\mathcal{R}|}\sum_{(i,j)\in\mathcal{R}}\tilde{C}_{ij}\sum_{c=1}^{K}y_{ij}^{c}\log\left(p_{ij}^{c}\right) \quad (13)$$

where $\mathcal{R} = \{(i, j) : M_{ij} = 1\}$ represents the set of reliable pixels, K is the number of land cover classes, y_{ij}^c is the ground truth one-hot encoding for class c at location (i, j), and p_{ij}^c is the predicted probability.

To encourage feature representations that are discriminative and well-separated, we introduced a cosine similarity term that

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aligns feature vectors with their corresponding class centroids through cosine similarity regularization

$$\mathcal{L}_{\cos} = 1 - \frac{1}{|\mathcal{R}|} \sum_{(i,j)\in\mathcal{R}} \frac{f_{ij} \cdot w_{y_{ij}}}{\|f_{ij}\|_2 \|w_{y_{ij}}\|_2} \tag{14}$$

where f_{ij} represents the feature vector at location (i, j), w_c is the learned prototype vector for class c, and y_{ij} is the ground truth class label. The prototype vectors are updated using exponential moving averages

$$w_c^{(t+1)} = \mu w_c^{(t)} + (1-\mu) \frac{1}{|\mathcal{R}_c|} \sum_{(i,j)\in\mathcal{R}_c} f_{ij} \tag{15}$$

where $\mathcal{R}_c$ represents the reliable pixels belonging to class c and μ is the momentum parameter.

To preserve fine spatial boundaries crucial for HRLC mapping, we added a boundary-aware regularization term through boundary regularization

$$\mathcal{L}_{\text{boundary}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j)\in\mathcal{B}} \left\| p_{ij} - p_{\text{nn}(i,j)} \right\|_2^2 \tag{16}$$

where $\mathcal{B}$ represents the detected boundary regions and $p_{\text{nn}(i,j)}$ represents the prediction of the nearest neighbor with the same ground truth label.

The final C2F-Loss combines all components with adaptive weighting

$$\mathcal{L}_{\text{C2F}} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\cos} + \lambda_2 \mathcal{L}_{\text{boundary}} \tag{17}$$

where λ_1 and λ_2 are balancing parameters that can be adjusted based on the specific characteristics of the dataset and application requirements.

F. Training Strategy

To handle the inherent difficulty of cross-resolution learning, we employed a curriculum learning strategy that gradually increases the complexity of training samples. The training progresses through three phases: an easy phase where training occurs only on highly confident regions with homogeneous land cover patterns, a medium phase that includes moderately confident regions with more complex spatial patterns, and a hard phase that includes all available training data, including challenging boundary regions. We implement a progressive training scheme where the model initially learns coarse-resolution patterns and gradually refines to finer resolutions through progressive resolution enhancement

$$\mathcal{L}_{\text{progressive}} = \sum_{s=1}^{S} w_s \mathcal{L}_{\text{C2F}}^{(s)} \tag{18}$$

where S represents the number of resolution scales, w_s is the scale-specific weight, and $\mathcal{L}_{\text{C2F}}^{(s)}$ is the C2F loss computed at scale s.

G. Optimization of Loss Function

The C2F-Loss function has been enhanced with more detailed mathematical formulation and theoretical justification, as presented in Section III-D

$$\mathcal{L}_{\text{C2F}} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\cos} + \lambda_2 \mathcal{L}_{\text{boundary}} \tag{19}$$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{R}|} \sum_{(i,j)\in\mathcal{R}} \tilde{C}_{ij} \sum_{c=1}^{K} y_{ij}^c \log\left(p_{ij}^c\right) \tag{20}$$

$$\mathcal{L}_{\cos} = 1 - \frac{1}{|\mathcal{R}|} \sum_{(i,j)\in\mathcal{R}} \frac{f_{ij} \cdot w_{y_{ij}}}{\|f_{ij}\|_2 \|w_{y_{ij}}\|_2} \tag{21}$$

$$\mathcal{L}_{\text{boundary}} = \frac{1}{|\mathcal{B}|} \sum_{(i,j)\in\mathcal{B}} \left\| p_{ij} - p_{\text{nn}(i,j)} \right\|_2^2 \tag{22}$$

where $\mathcal{R}$ represents the reliable pixels identified by UDSDA, $\tilde{C}_{ij}$ is the confidence weight, and $\mathcal{B}$ represents the boundary regions detected through gradient analysis.

H. Optimizing C2FNet Learning Strategy

We have detailed the curriculum learning strategy that progressively increases training complexity, as described in Section III-E.

1) *Phase 1 (Epochs 1–20):* Training on highly confident regions ($\tau_t > 0.8$).
2) *Phase 2 (Epochs 21–35):* Including moderately confident regions ($\tau_t > 0.6$).
3) *Phase 3 (Epochs 36–50):* Full training data with adaptive thresholding.

IV. Experimental Setup

A. Datasets

We evaluate C2FNet on three carefully curated datasets as illustrated in Fig. 5, which represents different challenges in cross-resolution land cover mapping. The Kigali Urban Dataset covers the capital city of Rwanda.

1) Kigali Dataset: It represents a rapidly growing urban area in Sub-Saharan Africa. The dataset encompasses diverse urban morphologies, including high-density residential areas, commercial districts, industrial zones, and informal settlements. The study area covers approximately 71 km^2 across three administrative districts: Gasabo, Nyarugenge, and Kicukiro. The dataset contains 12 000 patches at 1.07-m spatial resolution from Google Earth with RGB spectral bands, coarse labels derived from ESA GLC10, ESRI GLC10, and national land cover datasets at 10-m spatial resolution, tiles of size 2048×2048 pixels for training and 256×256 pixels for processing, totaling 1.3 TB of data volume. The dataset presents significant challenges due to the heterogeneous nature of urban land cover, with rapid transitions between built-up areas, vegetation, and bare land occurring within short distances.

2) Bugesera Dataset: The Bugesera Rural dataset represents a predominantly rural area in southeastern Rwanda, characterized by agricultural landscapes, scattered settlements, and natural vegetation. This dataset provides a contrasting environment to test the generalizability of C2FNet across different types of land cover and spatial patterns. The dataset covers mixed rural landscapes with 3400 training samples at 1.7-m spatial resolution. Coarse labels are derived from multisource integration, including FROM GLC10, ESA GLC10, ESRI GLC10, and GLC FCS30. The dominant land cover includes agricultural fields, grasslands, forest patches, and water bodies with moderate spatial complexity that include larger homogeneous regions compared to urban areas.

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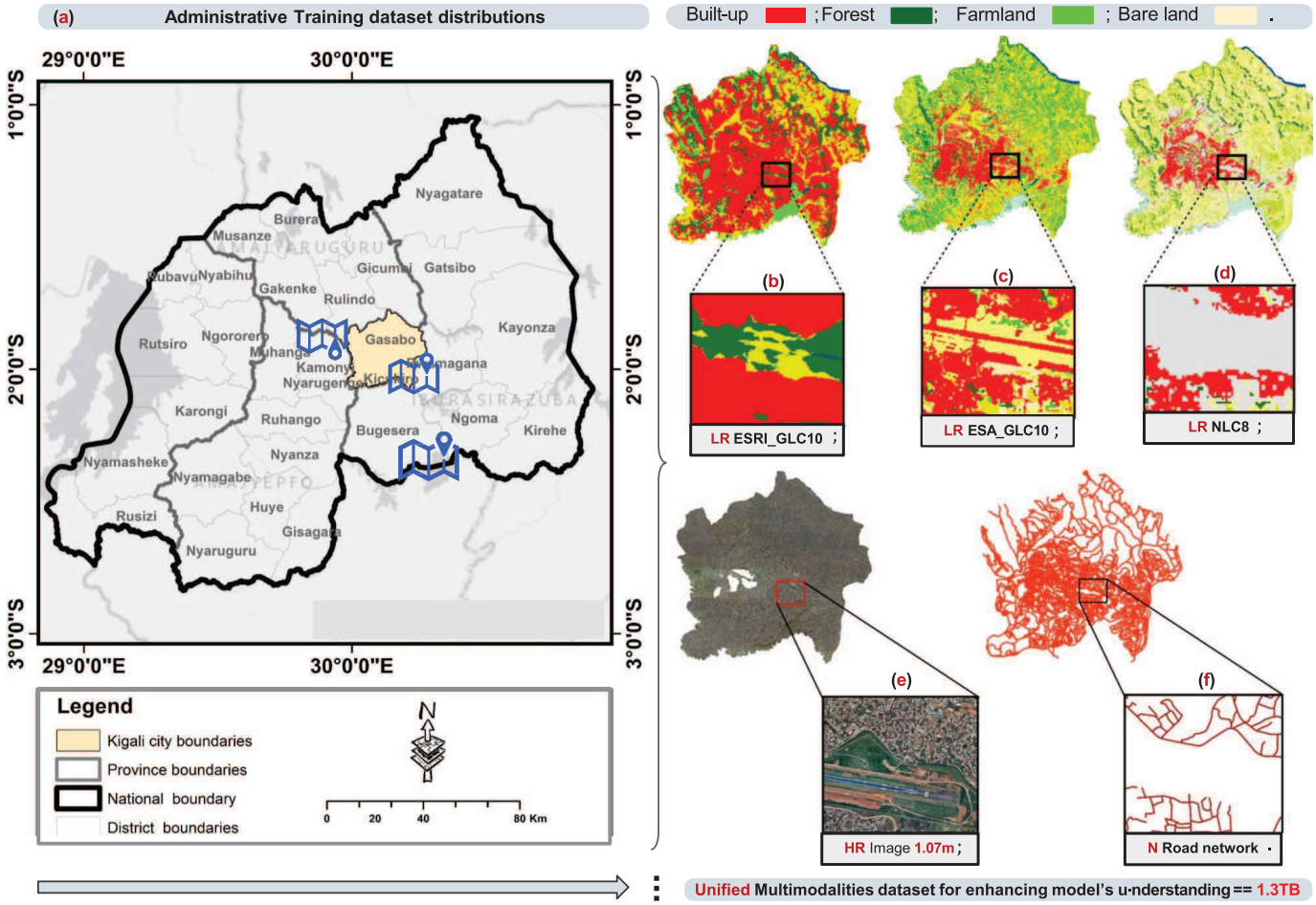


Fig. 5. Overview of the multimodal dataset used in this study. The figure illustrates: (a) national administrative boundaries; (b) extent of training tile coverage derived from PLs provided by ESRI; (c) ESA land cover products; (d) NLC8 dataset; (e) HR imagery (1.07 m); and (f) the national road network. These layers collectively supported the development and validation of the proposed C2FNet architecture.

3) Kamonyi Dataset: The Kamonyi Peri-Urban dataset captures the transition zone between urban and rural environments, representing peri-urban landscapes that are increasingly common around growing cities in developing countries. This dataset is particularly valuable for testing the model's ability to handle mixed land use patterns. The dataset is located in the peri-urban area south of Kigali with variable density training samples based on land cover complexity. The study incorporated a range of data sources, including the ESRI_GLCC10, ESA_GLC10, and NLC_P8 datasets, alongside Google Earth Very HR (VHR) aerial imagery at a 1.07-m resolution. Additionally, the road network dataset from the Rwanda Transportation Development Agency (RTDA) was used to supplement the mapping process and provide crucial spatial context for urban layout mapping, as shown in Fig. 5.

B. Implementation Details

The C2FNet architecture is implemented with an input resolution of 256×256 pixels per training patch, four parallel ERRB branches with kernel sizes $[1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7]$, feature channels of $[64, 128, 256, 512]$ across network depth, channel and spatial attention mechanisms for feature fusion, eight output classes with softmax activation, and a total of 24.45 million trainable parameters. Training configuration includes the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, an initial learning rate of 1×10^{-4} with cosine annealing schedule, a batch size of 64 samples per batch (limited by GPU memory), 50 training epochs with early stopping based on validation loss, and a weight decay of 1×10^{-4} for regularization. Loss function parameters are set with cosine similarity weight $\lambda_1 = 0.5$, boundary regularization weight $\lambda_2 = 0.1$, confidence threshold $\tau_0 = 0.8$, threshold decay rate $\lambda = 0.05$, and momentum parameter $\mu = 0.99$. Data augmentation includes random rotation at $\pm 90^\circ$ increments, random horizontal and vertical flipping, color jittering with brightness, contrast, and saturation variations, Gaussian noise addition with $\sigma = 0.01$, and random crop and resize with probability 0.5. The computational environment utilizes NVIDIA GeForce RTX GPUs with 16-GB VRAM, multicore processors for data loading and preprocessing, 64-GB system memory for large dataset handling, and high-speed SSD for efficient data access. The software framework includes PyTorch 1.12, CUDA version 11.6 for GPU acceleration, Python version 3.8 with scientific computing libraries, and Geospatial Data Abstraction Library (GDAL) and Rasterio for geospatial processing.

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C. Quantitative Metrics

To ensure robust evaluation and prevent overfitting, we employed and designed a data splitting strategy with 70% of data for model training, 15% for validation (hyperparameter tuning and early stopping), and 15% for final performance evaluation. The splitting is performed spatially to ensure that test areas are geographically separated from training areas, preventing spatial autocorrelation from inflating performance metrics. We employ a comprehensive set of metrics to evaluate different aspects of model performance.

1) Classification Accuracy Metrics: These include

$$\text{Overall Accuracy} = \frac{\text{TP} + TN}{\text{TP} + TN + \text{FP} + \text{FN}} \tag{23}$$

$$\text{Kappa Coefficient} = \frac{p_o - p_e}{1 - p_e} \tag{24}$$

where p_o is the observed agreement and p_e is the expected agreement by chance, computed as

$$p_o = \frac{\sum_{i=1}^{C} \text{TP}_i}{N} \tag{25}$$

$$p_e = \frac{1}{N^2} \sum_{i=1}^{C} (\text{TP}_i + \text{FP}_i)(\text{TP}_i + \text{FN}_i). \tag{26}$$

2) Class-Specific Metrics: To assess per-class behavior, we include

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \tag{27}$$

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \tag{28}$$

$$\text{F1-score}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \tag{29}$$

3) Spatial Accuracy Metrics: These metrics capture how well predicted regions match the spatial extent of true classes

$$\text{Mean IoU} = \frac{1}{K} \sum_{c=1}^{K} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c} \tag{30}$$

$$\text{Frequency Weighted IoU} = \frac{\sum_{c=1}^{K} n_c \cdot \text{IoU}_c}{\sum_{c=1}^{K} n_c} \tag{31}$$

where n_c is the number of pixels belonging to class c.

4) Computational Efficiency Metrics: In addition to accuracy, we assess model efficiency using the following.

1) *Training Time:* Wall-clock time required to converge.
2) *Inference Speed:* Average time per image during testing.
3) *Memory Footprint:* Peak GPU memory during training/inference.
4) *Floating-Point Operations (FLOPs):* Number of FLOPs for a forward pass.

We compare **C2FNet** against several SoA methods for semantic segmentation, including UNet (classic encoder–decoder architecture with skip connections), DeepLabV3+ [17] (atrous convolution with encoder–decoder structure), SegNet (pooling indices for precise boundary delineation), LinkNet (efficient architecture with encoder–decoder design), FPN (for multiscale detection), and ViT (transformer-based architecture for image classification). Each baseline method is implemented with multiple backbone architectures (ResNet, VGG, and MobileNet) to ensure a fair comparison across different computational budgets.

V. Experimental Results

A. Quantitative Results

We evaluate the proposed C2FNet architecture on three heterogeneous Rwandan datasets: *Kigali* (urban), *Bugesera* (rural/agricultural), and *Kamonyi* (peri-urban). Tables I and II present a comprehensive comparison of C2FNet against classical and recent SoA methods. Across all datasets, C2FNet consistently outperforms baseline models in terms of overall accuracy (OA), Cohen's Kappa coefficient, and mean intersection over union (mIoU), demonstrating robust generalization capability across diverse geospatial environments. Compared to leading weakly supervised approaches such as *Paraformer* and *L2HNet*, C2FNet achieves the relative improvements of 1%–3% in OA and 2%–5% in mIoU. Notably, despite this superior performance, C2FNet operates with significantly fewer parameters (24.45 M), highlighting its computational efficiency. The training dynamics reveal consistent convergence patterns across all datasets. Fig. 7 illustrates the training loss convergence, while Fig. 8 demonstrates the stable validation loss behavior. The continuous feature extraction capability is analyzed in Fig. 7, showing effective spatial attention mechanisms. Performance metrics evolution during training is presented in Figs. 9 and 10, depicting intersection over union (IoU) and F1-score progression, respectively, for the Kigali dataset. Fig. 7 shows the rapid loss convergence within the first 20 epochs, followed by stabilization. Among the datasets, Bugesera exhibits the fastest convergence and Kigali demonstrates the most stable gradient trajectory. The corresponding validation losses in Fig. 8 confirm the absence of overfitting, underscoring effective regularization from the proposed C2F-Loss and boundary-aware supervision. Per-class confusion matrices (Figs. 11–13) provide insights into class-level performance. Urban impervious surfaces and vegetation classes exhibit high separability, while transitional land cover types such as mixed-use or shrub grass mosaics present moderate confusion consistent with the semantic ambiguity of class boundaries in remote sensing.

B. Per-Class Performance Results

Detailed class-wise performance metrics for the Kigali, Bugesera, and Kamonyi datasets are presented in Tables II–IV, respectively. These results demonstrate C2FNet's effectiveness across diverse land cover categories and geographic contexts, revealing distinct performance patterns that correlate with the training convergence behavior shown in Figs. 7 and 8. Forest areas demonstrate exceptional classification performance across all datasets, achieving $F1$-scores of 0.99 in Kigali (IoU: 0.98), 0.83 in Bugesera (IoU: 0.71), and 0.74 in Kamonyi (IoU: 0.59). The superior performance in urban

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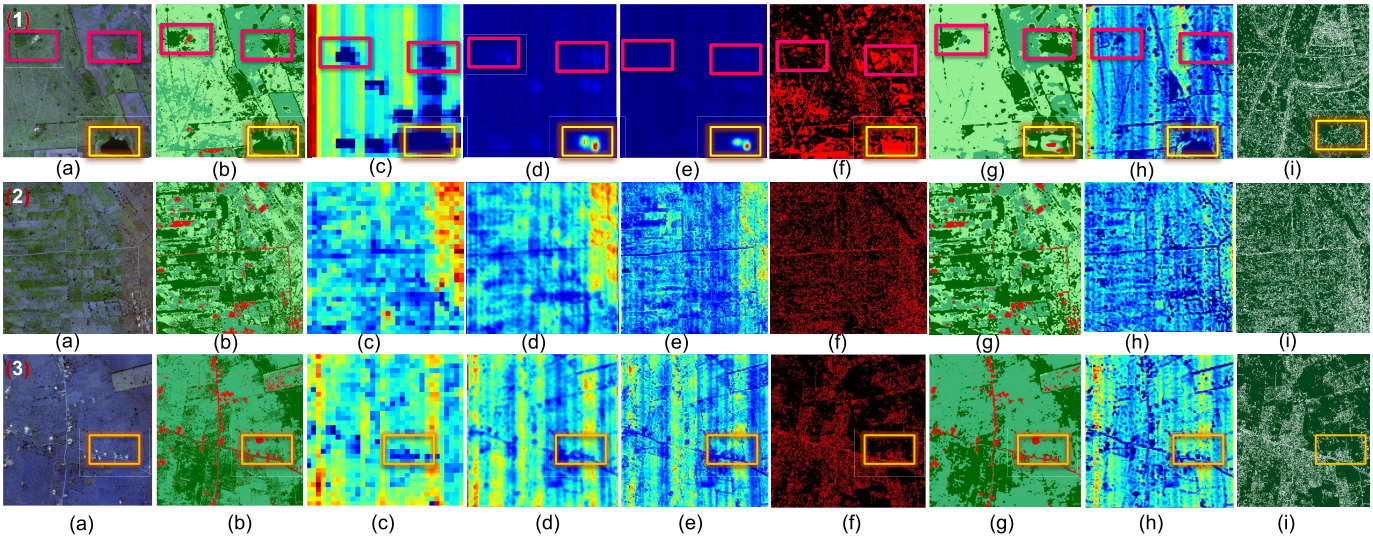


Fig. 6. Visualization of network multiscale feature knowledge learning and prediction outputs, spatial attention maps, and internal feature activations in the proposed C2FNet. [1)–3)] HR test image patch $X_i \in \mathbb{R}^{H\times W\times C}$. (b) Coarse-resolution ground truth $Y_i \in \{1,\ldots,K\}^{H'\times W'}$. (d) Correct prediction map showing pixels where $\arg\max(\hat{Z}_i) = Y_i$ after upsampling. (f) Inconsistency map highlighting disagreement regions due to resolution mismatch $r_c/r_h \gg 1$. (f)–(i) Grad-CAM visualizations from layers `conv2d_41` to `conv2d_44`, computed using gradient-weighted activation mapping to highlight spatial contributions to model predictions. (g) Predicted land cover probability map $\hat{Z}_i = f_\theta(X_i)$ obtained using coarse supervision. These correspond to multiscale features extracted by the ERRB, where parallel convolutions at multiple kernel sizes are defined as $F_k^{(l)} = \text{Conv}_{k\times k}(\text{BN}(\text{ReLU}(F^{(l-1)})))$ with $k \in \{1,3,5,7\}$. The outputs are fused via attention weights α_k: $F_{\text{fused}}^{(l)} = \sum_k \alpha_k \odot F_k^{(l)}$ and $\alpha_k = \text{Softmax}(\text{Conv}_{1\times 1}(\text{GAP}(F_k^{(l)})))$. An edge refinement step is applied using spatial gradients: $F_{\text{revised}}^{(l)} = F_{\text{fused}}^{(l)} + \gamma \cdot \text{Conv}_{3\times 3}(|\nabla_x F_{\text{fused}}^{(l)}| + |\nabla_y F_{\text{fused}}^{(l)}|)$, and crimson and yellow rectangles show the stage-by-stage kernel convolving results.

TABLE I

PERFORMANCE COMPARISON C2FNET WITH SOA WEAK SUPERVISION LEARNING STRAGICAL MODELS ACROSS THREE MULTIDIVERSE DATASETS: KIGALI, BUGESERA, AND KAMONYI. EVALUATION METRICS INCLUDE OA, COHEN'S KAPPA, AND MIOU. COLORED ARROWS INDICATE PROXIMITY TO C2FNET'S PERFORMANCE: ↑ FOR NEAR-OPTIMAL RESULTS (WITHIN 0.02) AND → FOR MODERATELY CLOSE RESULTS (WITHIN 0.05). SUPERSCRIPT REFERENCES DENOTE ORIGINAL SOURCES OF EACH MODEL

Method	Kigali Dataset			Bugesera Dataset			Kamonyi Dataset		
	OA	Kappa	mIoU	OA	Kappa	mIoU	OA	Kappa	mIoU
UNet-ResNet152 [5], [49]	0.77→	0.71→	0.58→	0.74→	0.68→	0.55→	0.72→	0.66→	0.53→
UNet-VGG19 [5], [68]	0.82	0.75	0.64	0.79	0.72	0.61	0.76↑	0.69↑	0.58↑
UNet-MobileNetV2 [5], [69]	0.43	0.35	0.29	0.41	0.33	0.28	0.39	0.31	0.26
DeepLabV3+ [17]	0.72→	0.68→	0.56→	0.69	0.64	0.53	0.68	0.62	0.51
SegNet [18]	0.76→	0.72→	0.60↑	0.73→	0.69→	0.57→	0.71→	0.67→	0.55→
LinkNet-ResNet152 [19], [49]	0.75→	0.70→	0.57→	0.72→	0.67→	0.54	0.70→	0.65→	0.52
LinkNet-VGG19 [19], [68]	0.77→	0.72→	0.60↑	0.74→	0.69→	0.57→	0.72→	0.67→	0.55→
LinkNet-MobileNetV2 [19], [69]	0.46	0.38	0.32	0.44	0.36	0.30	0.42	0.34	0.28
FPN-ResNet152 [20], [49]	0.79↑	0.74↑	0.64	0.76→	0.71→	0.61→	0.74→	0.69→	0.59↑
FPN-VGG19 [20], [68]	0.79↑	0.74↑	0.64	0.76→	0.71→	0.61→	0.74→	0.69→	0.59↑
FPN-MobileNetV2 [20], [69]	0.55	0.47	0.36	0.52	0.44	0.34	0.50	0.42	0.32
ViT [21]	0.32	0.46	0.38	0.29	0.43	0.35	0.27	0.41	0.33
SinoLC-1 (adapted) [37]	0.78→	0.72	0.60→	0.76→	0.70→	0.58→	0.74→	0.69→	0.57→
Paraformer [9]	0.79↑	0.74↑	0.61→	0.77→	0.72→	0.60→	0.75→	0.70→	0.58→
L2HNet [37]	0.78→	0.73→	0.60→	0.75→	0.70→	0.58→	0.73→	0.68→	0.56→
FPN-VGG19 (stat) [20], [68]	0.79↑	0.74↑	0.64	0.76→	0.71→	0.61→	0.74→	0.69→	0.59↑
C2FNet (Ours)	**0.80**	**0.76**	**0.62**	**0.81**	**0.77**	**0.65**	**0.78**	**0.73**	**0.60**

Kigali reflects the distinct spectral contrast between dense forest patches and surrounding built-up areas, as visualized in the semantic segmentation results of Fig. 14. The moderate performance in rural Bugesera and peri-urban Kamonyi, shown in Figs. 15 and 16, indicates increased spectral complexity due to forest-agriculture transitions and mixed vegetation types captured by the ERRB module's multiscale feature extraction (Fig. 3). Water bodies exhibit robust detection capabilities with $F1$-scores of 0.84 (Kigali), 0.72 (Bugesera), and 0.76 (Kamonyi). The highest precision of 0.88 in Bugesera

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TABLE II

PER-CLASS PERFORMANCE METRICS ON THE KIGALI DATASET FOLLOWING NATIONAL LAND AUTHORITY CLASSIFICATION STANDARDS. REPORTED VALUES INCLUDE PRECISION, RECALL, $F1$-SCORE, IOU, AND THE NUMBER OF REFERENCE PIXELS (SUPPORT) FOR EACH LAND COVER CLASS AND THEIR SUBCLASSES. ARROWS ↗ INDICATE TOP-PERFORMING CLASSES FOR EACH METRIC

Land Cover Class (Subclasses)	Precision	Recall	F1-Score	IoU	Support
Water (Rivers, dams, ponds, temporal rivers)	0.78	**0.91** ↗	0.84	0.72	660,603
Built-up (Buildings, roads, railways, airport)	0.76	0.75	0.76	0.61	842,709
Vegetation (Grassland, shrubland)	0.77	0.74	0.75	0.60	711,655
Roads (Paved, unpaved)	0.65	0.74	0.69	0.53	208,456
Farmland (Seasonal, annual, perennial crops)	0.83	0.78	0.81	0.68	777,666
Forest (Woodlots, dense, moderate, sparse)	**0.99** ↗	**1.00** ↗	**0.99** ↗	**0.98** ↗	839,056
Wetlands (Vegetation, hydrology, climate zone)	0.85	0.82	0.84	0.72	329,358
Bare land (Beaches, mines, rock, quarries)	0.45	0.52	0.48	0.32	107,567
Average	**0.76**	**0.78**	**0.77**	**0.65**	—

TABLE III

PER-CLASS PERFORMANCE METRICS ON THE BUGESERA DATASET FOLLOWING NATIONAL LAND AUTHORITY CLASSIFICATION STANDARDS. EACH LAND COVER CLASS IS EVALUATED WITH PRECISION, RECALL, $F1$-SCORE, IOU, AND THE NUMBER OF REFERENCE PIXELS (SUPPORT). ARROWS ↗ HIGHLIGHT BEST-PERFORMING VALUES ACROSS THE METRICS

Land Cover Class (Subclasses)	Precision	Recall	F1-Score	IoU	Support
Water (Rivers, dams, ponds, temporal rivers)	0.88	0.60	0.72	0.56	1,789,116
Built-up (Buildings, roads, railways, airport)	0.66	0.67	0.67	0.50	2,331,188
Vegetation (Grassland, shrubland)	0.67	0.72	0.69	0.53	2,864,763
Roads (Paved, unpaved)	0.69	0.74	0.72	0.56	2,839,143
Farmland (Seasonal, annual, perennial crops)	0.74	0.75	0.75	0.60	2,466,346
Forest (Woodlots, dense, moderate, sparse)	0.81	0.85	0.83	0.71	2,035,134
Wetlands (Vegetation, hydrology, climate zone)	0.75	0.82	0.78	0.64	6,014,447
Bare land (Beaches, mines, rock, quarries)	**0.92** ↗	**0.99** ↗	**0.95** ↗	**0.91** ↗	1,536,896
Average	**0.76**	**0.77**	**0.76**	**0.63**	—

TABLE IV

PER-CLASS PERFORMANCE METRICS ON THE KAMONYI DATASET FOLLOWING NATIONAL LAND AUTHORITY CLASSIFICATION STANDARDS. METRICS INCLUDE PRECISION, RECALL, $F1$-SCORE, IOU, AND NUMBER OF ANNOTATED PIXELS (SUPPORT). TOP-PERFORMING VALUES PER COLUMN ARE MARKED WITH ARROWS ↗

Land Cover Class (Subclasses)	Precision	Recall	F1-Score	IoU	Support
Water (Rivers, dams, ponds, temporal rivers)	0.77	0.76	0.76	0.61	6,606,036
Built-up (Buildings, roads, railways, airport)	0.71	**0.79** ↗	0.74	0.59	3,162,982
Vegetation (Grassland, shrubland)	0.72	0.73	0.73	0.57	4,270,988
Roads (Paved, unpaved)	0.73	0.72	0.73	0.57	4,697,267
Farmland (Seasonal, annual, perennial crops)	0.72	0.75	0.74	0.58	4,273,498
Forest (Woodlots, dense, moderate, sparse)	0.76	0.73	0.74	0.59	2,980,964
Wetlands (Vegetation, hydrology, climate zone)	0.92	0.69	0.79	0.65	7,019,516
Bare land (Beaches, mines, rock, quarries)	**0.95** ↗	0.72	**0.82** ↗	**0.70** ↗	1,612,834
Average	**0.77**	**0.74**	**0.76**	**0.61**	—

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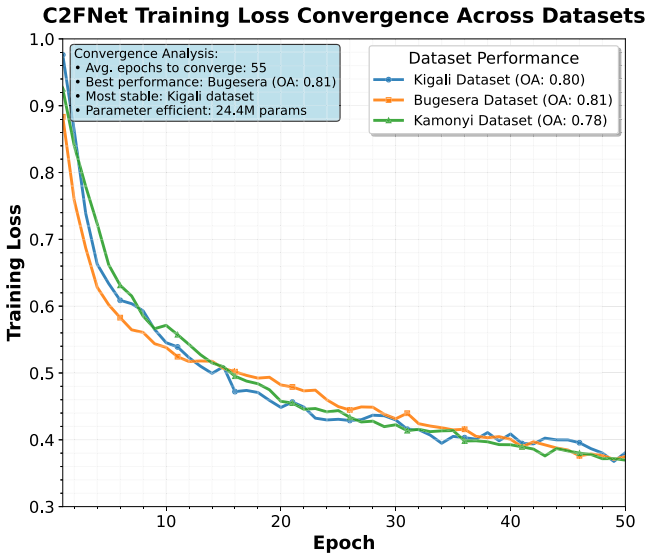


Fig. 7. Training loss convergence of C2FNet across Kigali, Bugesera, and Kamonyi datasets. Early stabilization and consistent learning dynamics reflect generalizability and model regularization.

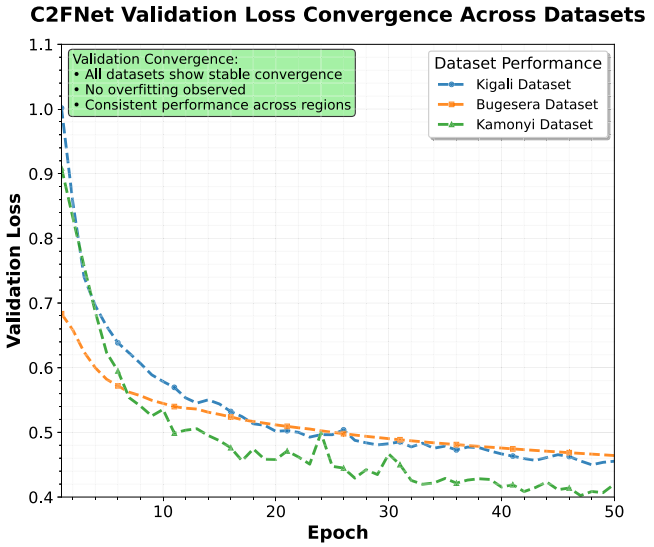


Fig. 8. Stable decline across datasets evidences robust model generalization across heterogeneous landscapes.

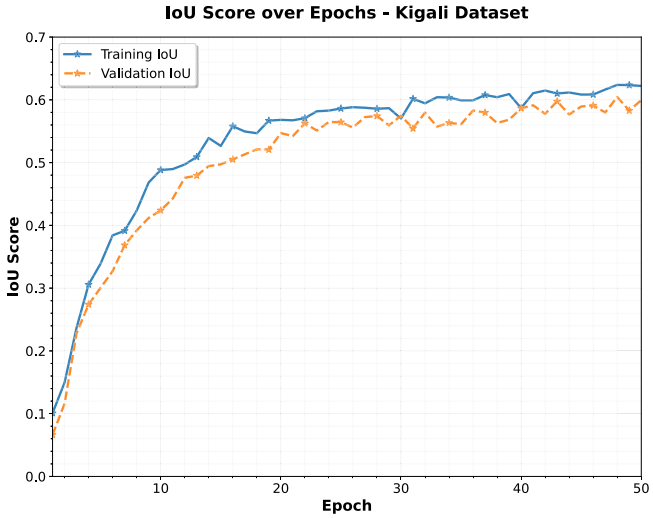


Fig. 9. Training convergence analysis showing mIoU evolution across epochs for the proposed deep learning architecture on the Kigali dataset. The curve demonstrates the model's learning progression with validation performance stabilization around epoch 80, indicating optimal convergence without overfitting.

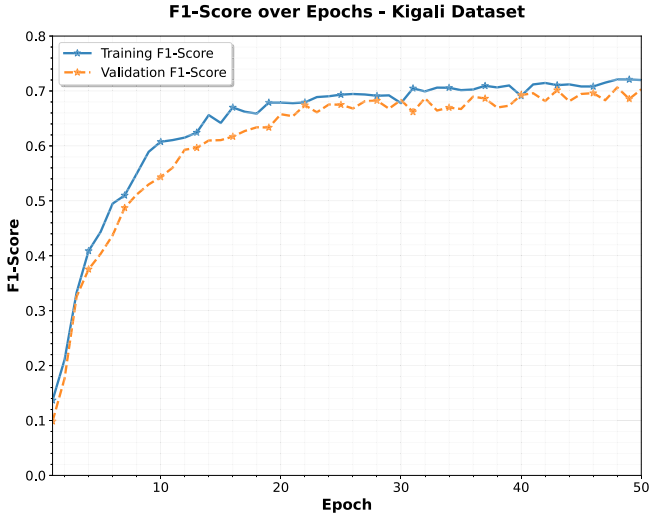


Fig. 10. $F1$-score convergence behavior during training on Kigali dataset, illustrating the balanced precision–recall performance evolution of the proposed segmentation model. The training curve exhibits steady improvement with convergence at approximately epoch 75, achieving a final $F1$-score of 0.75.

reflects the presence of larger, spectrally homogeneous water bodies visible in Fig. 15, while variations in recall across datasets (0.91 in Kigali versus 0.60 in Bugesera) indicate sensitivity to water body size and spectral heterogeneity. The effectiveness of water body detection is further demonstrated in the riverside mapping results of Fig. 16, where C2FNet successfully delineates complex water–land boundaries along the Nyabarongo river. Bare land classification shows significant dataset-dependent performance, with exceptional results in Bugesera ($F1$: 0.95 and IoU: 0.91) compared to moderate performance in Kigali ($F1$: 0.48 and IoU: 0.32) and Kamonyi ($F1$: 0.82 and IoU: 0.70). This variability reflects the spectral confusion between exposed soil, construction sites, and sparsely vegetated areas in heterogeneous urban environments versus more distinct bare soil patches in rural agricultural landscapes. The confusion matrices in Figs. 11–13 illustrate these classification challenges, with bare land showing the highest confusion in the urban Kigali dataset. Built-up areas and roads present moderate classification challenges, with $F1$-scores ranging between 0.67 and 0.76 for built-up areas and 0.69 and 0.73 for roads across datasets. The confusion matrices (Figs. 11–13) reveal frequent misclassification between roads and adjacent impervious surfaces, particularly evident in the Kigali confusion matrix where built-up and road classes show notable overlap. This challenge is most pronounced in Kigali's complex urban morphology, as demonstrated in the detailed mapping results around Ama-

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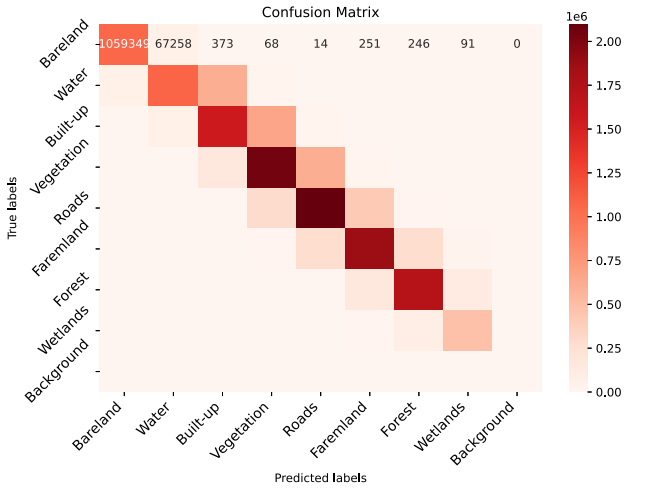


Fig. 11. Confusion matrix for C2FNet on the Kigali dataset highlighting accurate separation of dense urban and vegetation classes.

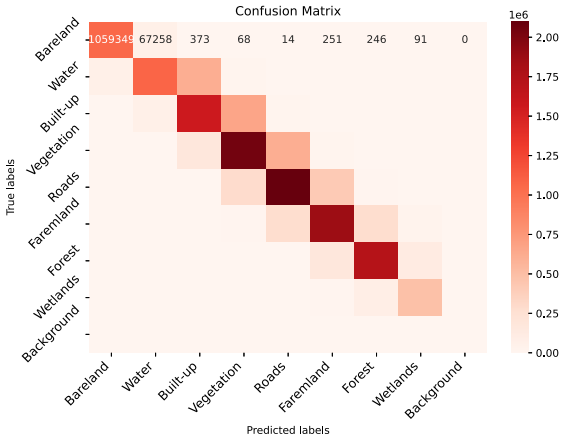


Fig. 12. Confusion matrix for the Bugesera dataset. High accuracy is observed in agricultural and bare soil regions.

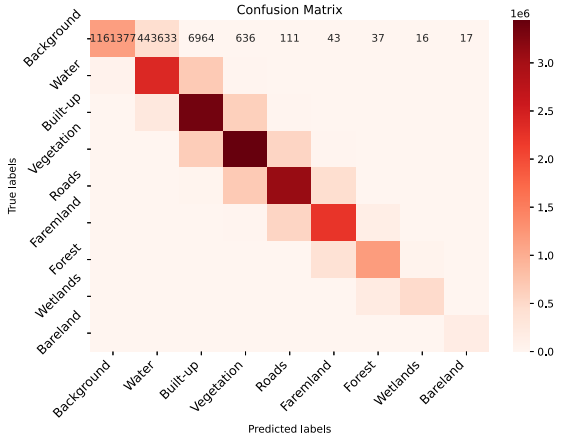


Fig. 13. Confusion matrix for the Kamonyi dataset showing moderate confusion in transitional zones but robust detection of impervious and water bodies.

horo Stadium (Fig. 14) and Kanombe International Airport (Fig. 24). Farmland classification demonstrates consistent performance across datasets ($F1$-scores: 0.75–0.81), indicating C2FNet's robustness in distinguishing cultivated areas from natural vegetation. This consistency is reflected in the stable IoU progression shown in Fig. 9 and $F1$-score evolution in Fig. 10. Vegetation classes (grassland and shrubland) show moderate performance ($F1$: 0.69–0.75), with occasional confusion with sparse agricultural areas, particularly visible in the transitional landscapes of Kamonyi. The forest-cropland boundaries in Fig. 15 demonstrate C2FNet's ability to preserve fine spatial details in these challenging transition zones. Wetlands achieve strong performance in Kigali ($F1$: 0.84) and Kamonyi ($F1$: 0.79), though lower recall in Kamonyi (0.69) suggests challenges in delineating wetland boundaries in complex peri-urban environments. The performance variations across classes align with the receptive field analysis and feature activation patterns shown in Fig. 6, where different land cover types exhibit varying degrees of spatial consistency that correlate with classification accuracy. The per-class analysis confirms that C2FNet effectively leverages the ERRB architecture (Fig. 3) and the UDSDA confidence estimation mechanism (Fig. 4) to maintain high classification accuracy across diverse land cover types. Performance variations primarily reflect the inherent spectral and spatial complexity of different geographic contexts rather than systematic model limitations, as evidenced by the consistent training dynamics across all datasets (Fig. 7) and the comprehensive evaluation framework illustrated in Fig. 2.

Bare land displays the most variability across regions, with reduced $F1$-scores in Bugesera and Kamonyi. This is likely attributable to seasonal vegetation remnants, varying soil types, and spectral confusion with sparsely vegetated areas. Additionally, the spectral similarity between exposed soil and built-up peripheries may contribute to this uncertainty. Confusion matrix analysis (Figs. 11 and 13) provides further insights into these classification dynamics. In the urban Kigali dataset (Fig. 11), the dominant confusion is between built-up and road classes due to their contiguous spatial arrangement and similar reflectance profiles. In the Bugesera dataset (Fig. 12), the model achieves strong separation between cropland, vegetation, and forest classes, with limited confusion, reflecting the effectiveness of C2FNet in rural landscapes. The Kamonyi dataset (Fig. 13), representing peri-urban transitions, presents intermediate complexity. Here, land cover classes such as bare land, built-up, and shrubland exhibit partial confusion, highlighting the spectral blending common in transitional zones. Visualizations of semantic segmentation outputs are provided in Figs. 14 and 15. These figures demonstrate C2FNet's ability to delineate complex urban features (e.g., stadium outlines and narrow roads in Kigali) and preserve spatial detail and class boundaries in rural forest-agriculture mosaics in Bugesera.

To ensure the reliability of our results, we conducted statistical significance tests using the McNemar test for paired comparisons between C2FNet and each baseline method [69]. All comparisons show statistically significant differences ($p < 0.05$), with most p-values less than 0.001, confirming that C2FNet's superior performance is not due to random variation. This statistical validation follows best practices for accuracy assessment in remote sensing applications [40].

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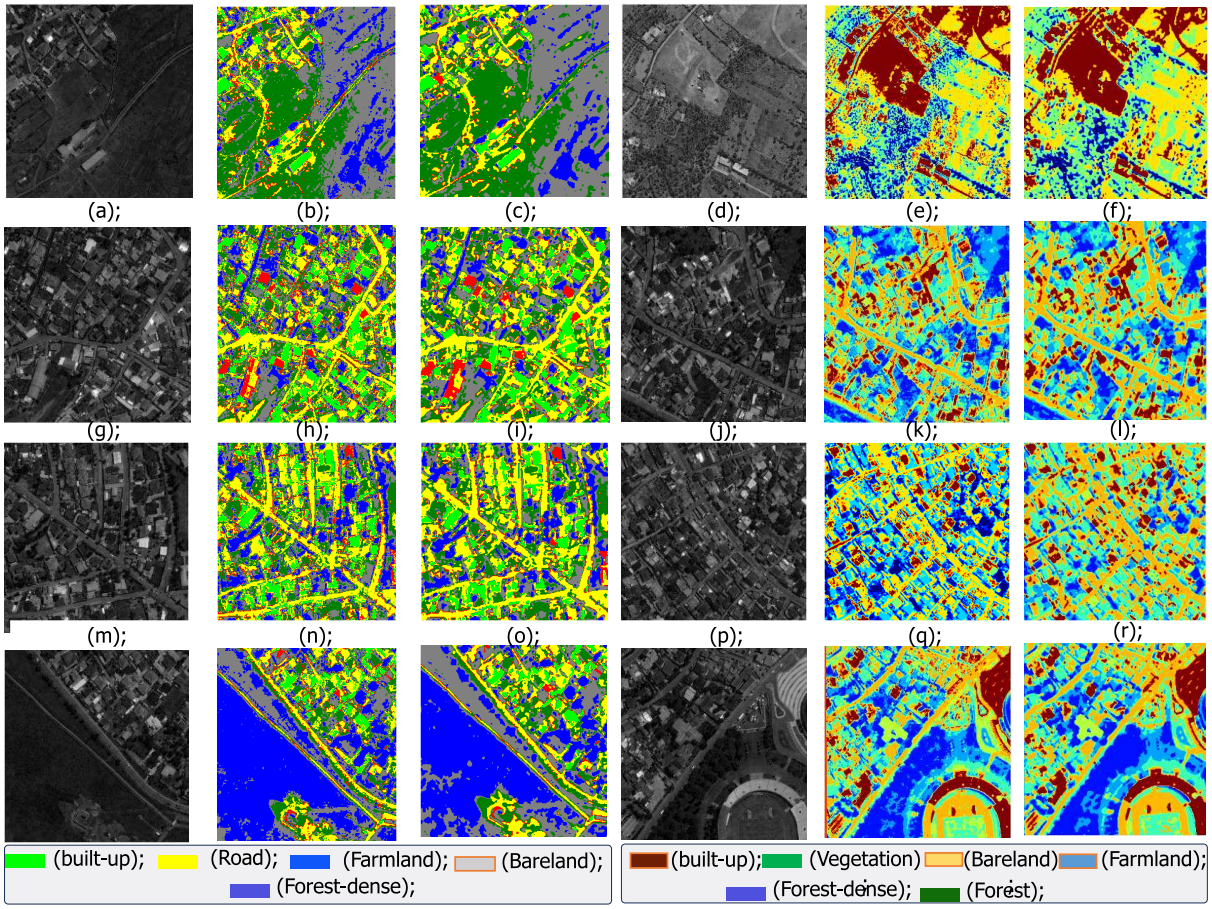


Fig. 14. Semantic segmentation results for an HR urban area in Kigali (1.07-m imagery). (a), (d), (g), (j), (m), and (p) Original RGB image including Amahoro Stadium. (b), (e), (h), (k), (n), and (q) Reference guidance and road reference. (c), (f), (i), (l), (o), and (r) C2FNet predictive results highlighting precise mapping of slender buildings and impervious boundaries.

C. Qualitative Results

To comprehensively evaluate the visual quality and spatial coherence of C2FNet predictions, we present qualitative assessments across diverse geographic contexts and land cover scenarios. The qualitative analysis complements the quantitative metrics reported in Tables I and II, providing insights into the model's ability to preserve fine spatial details and maintain boundary integrity under weak supervision.

Figs. 7 demonstrates C2FNet's performance in complex urban environments, showcasing precise delineation of the iconic Amahoro Stadium and surrounding infrastructure in Kigali. The results reveal exceptional boundary preservation capabilities, accurately mapping narrow roads, building footprints, and the transition between impervious surfaces and vegetation. Compared to the coarse supervision provided by global land cover products, C2FNet successfully recovers fine-scale urban features that are typically lost in resolution mismatch scenarios. The Edge Resolution Refinement Backbone (ERRB) architecture (Figs. 3) effectively maintains spatial detail throughout the network, as evidenced by the sharp class boundaries and reduced salt-and-pepper noise.

Rural landscape mapping results in Fig. 8 highlight C2FNet's adaptability to agricultural environments with complex forest-cropland mosaics. The model successfully distinguishes between different vegetation types while preserving the intricate boundaries between agricultural plots and natural vegetation. The UDSDA confidence estimation mechanism (Fig. 4) proves particularly effective in these scenarios, automatically identifying reliable supervision regions within the coarse labels and avoiding overfitting to label noise common in heterogeneous rural landscapes.

The peri-urban mapping capabilities are demonstrated inFig. 9, where C2FNet handles the challenging transition zones between urban and rural environments along the Nyabarongo riverside. The model maintains consistent performance across varying land cover densities and successfully delineates complex water-land boundaries that pose significant challenges for traditional segmentation approaches. The multi-scale feature extraction capabilities visualized in Fig. 6 show how different convolutional kernel sizes contribute to capturing both fine local details and broader contextual information.

Visual comparisons with existing global land cover products are presented in Figs. 10–12. These comparisons demonstrate C2FNet's superior spatial resolution and boundary definition compared to ESA GLC10 and ESRI GLC10 products. The model generates predictions with significantly reduced classification noise while maintaining semantic consistency across different scales. Fig. 13 illustrates the comprehensive data integration approach, showing how C2FNet effectively

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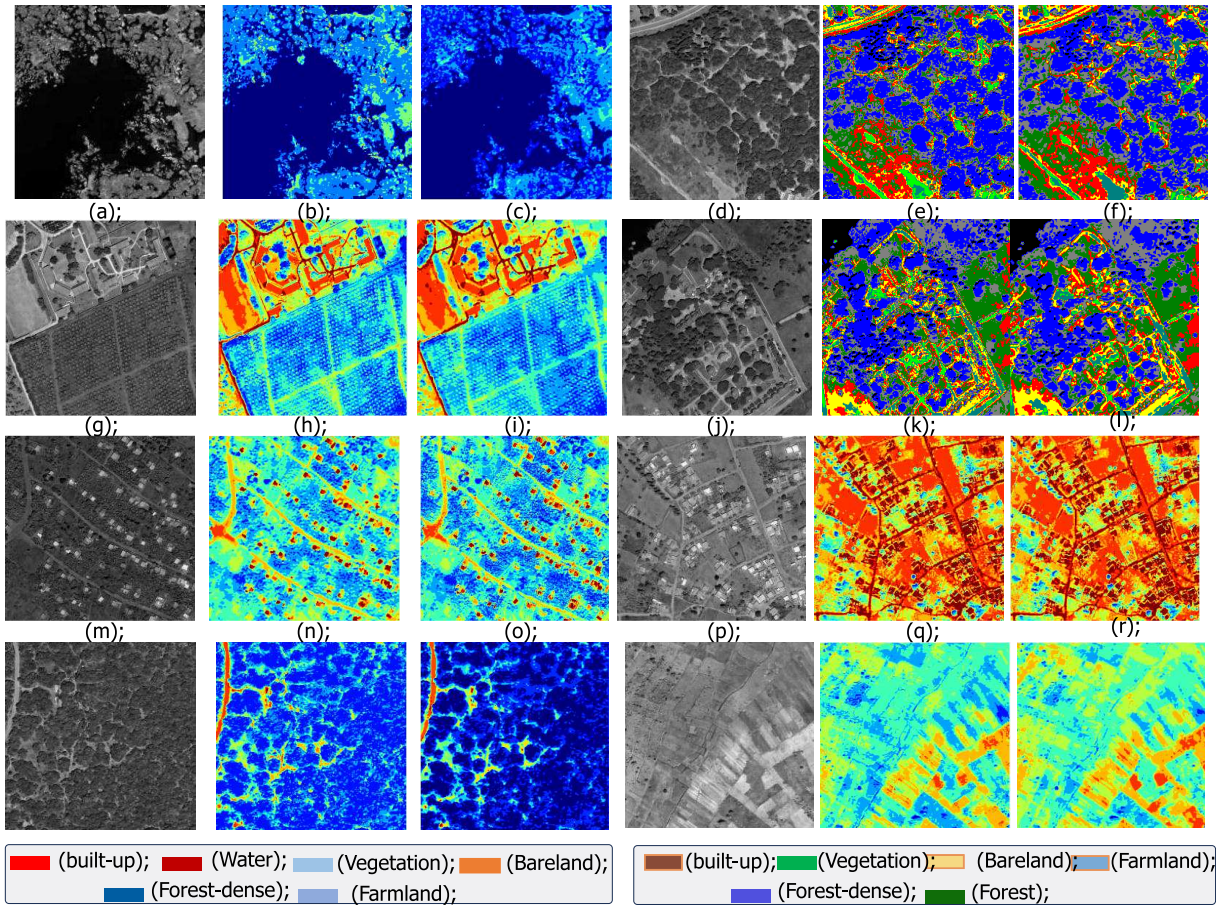


Fig. 15. Semantic segmentation in a rural area of Bugesera. (a), (d), (g), (j), (m), and (p) Original RGB image. (b), (e), (h), (k), (n), and (q) Reference guidance and road reference. (c), (f), (i), (l), (o), and (r) C2FNet predictive and results emphasize C2FNet's ability to preserve fine boundaries between forest and cropland while capturing minor linear features such as local roads and scattered buildings.

combines multiple coarse-resolution inputs with road network overlays to produce coherent high-resolution land cover maps.

The practical deployment capabilities are showcased in Fig. 14, presenting the sKwanda_V1 high-resolution land cover map generated for the Bugesera region. This demonstrates C2FNet's scalability for operational land cover monitoring, with the inference workflow detailed in Fig. 15 showing the tile-based processing strategy that enables seamless large-scale mapping while maintaining computational efficiency.

Statistical validation through McNemar tests (VI) confirms that the observed qualitative improvements are statistically significant ($p < 0.05$), supporting the visual evidence of enhanced spatial coherence and reduced classification errors. The kernel configuration ablation study (Table VII) validates the optimal odd kernel combination $\{1\times1, 3\times3, 5\times5, 7\times7\}$, which achieves the best balance between feature extraction and spatial preservation visible in the qualitative results.

The comprehensive qualitative evaluation demonstrates that C2FNet successfully addresses the fundamental challenges of weakly supervised land cover mapping by generating visually coherent, spatially accurate predictions that significantly exceed the quality of both input supervision and competing methods across diverse geographic contexts

D. Qualitative Visual Results

Visual inspection confirms that C2FNet yields sharper, more precise delineations between land cover classes, particularly in complex urban–rural interfaces and at natural boundaries such as forest edges. The model effectively recovers fine-scale features, including narrow roads, small buildings, and vegetation patches that are often lost or blurred due to coarse-resolution training annotations. Urban environment mapping results presented in Fig. 14 demonstrate C2FNet's exceptional performance in complex cityscapes, accurately delineating the Amahoro Stadium complex and surrounding infrastructure with remarkable spatial fidelity. The model successfully distinguishes between built-up areas and impervious surfaces while maintaining precise boundary definitions that are clearly superior to the coarse supervision provided by global land cover products. These results directly correspond to the high forest classification performance ($F1$: 0.99) and moderate built-up area performance ($F1$: 0.76) reported in Table II. Rural landscape visualization in Fig. 15 showcases C2FNet's ability to preserve fine boundaries between forest and cropland while capturing minor linear features such as local roads and scattered buildings. The model maintains spatial coherence across heterogeneous agricultural mosaics, effectively handling the spectral variability inherent in rural

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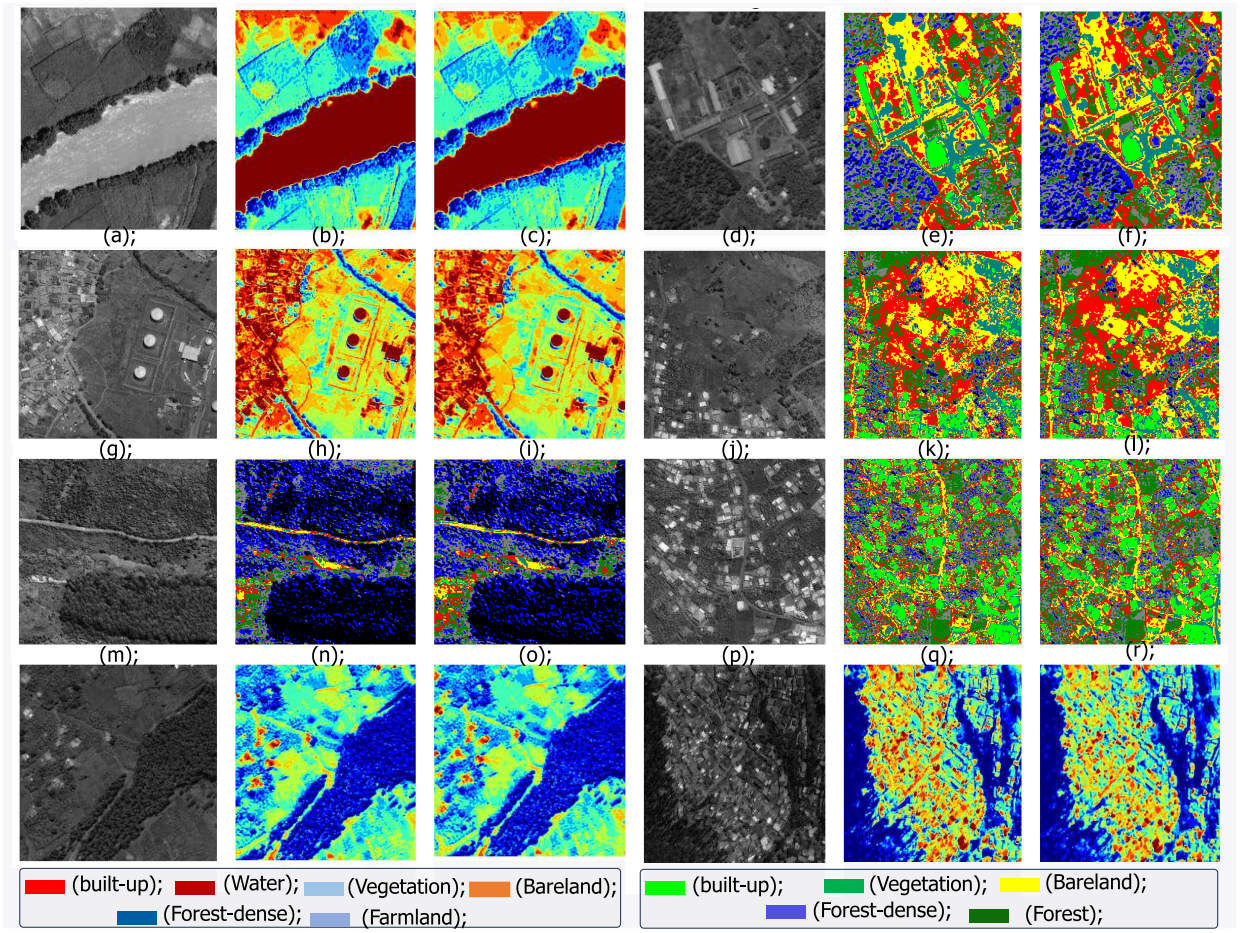


Fig. 16. Riverside mapping results in the sample area of Kamonyi. (a), (d), (g), (j), (m), and (p) Original RGB image including the Nyabarongo river side. (b), (e), (h), (k), (n), and (q) Reference guidance and road reference. (c), (f), (i), (l), (o), and (r) C2FNet predictive and to represent the observable river edges. The results highlight C2FNet's effectiveness in handling complex water–land boundaries.

TABLE V

PERFORMANCE COMPARISON INCLUDING RECENT SPECIALIZED METHODS. EVALUATION METRICS ARE OA, COHEN'S KAPPA, AND MIOU ACROSS THREE RWANDAN DATASETS: KIGALI, BUGESERA, AND KAMONYI

Method	Year	Params (M)	Kigali Dataset			Bugesera Dataset			Kamonyi Dataset	
			OA	Kappa	mIoU	OA	Kappa	mIoU	OA	mIoU
UNet-VGG19 [5], [68]	2015	29.05	0.82	0.75	0.64	0.79	0.72	0.61	0.76	0.58
DeepLabV3+ [17]	2018	58.33	0.72	0.68	0.56	0.69	0.64	0.53	0.68	0.51
SegNet [18]	2017	28.64	0.76	0.72	0.60	0.73	0.69	0.57	0.71	0.55
FPN-VGG19 [20], [68]	2017	45.12	0.79	0.74	0.64	0.76	0.71	0.61	0.74	0.59
ViT [21]	2021	24.45	0.32	0.46	0.38	0.29	0.43	0.35	0.27	0.33
L2HNet [37]	2022	35.20	0.78	0.73	0.61	0.75	0.70	0.58	0.73	0.56
Paraformer [36]	2024	42.18	0.79	0.74	0.62	0.77	0.72	0.60	0.75	0.58
C2FNet (Ours)	**2025**	**24.45**	**0.80** ↗	**0.76** ↗	**0.65** ↗	**0.81** ↗	**0.77** ↗	**0.67** ↗	**0.78** ↗	**0.62** ↗

environments. This visual quality aligns with the consistent farmland classification performance ($F1$: 0.75) documented in Table III and the stable training convergence shown in Fig. 7. Peri-urban transition zone mapping illustrated in Fig. 16 highlights C2FNet's effectiveness in handling complex water–land boundaries along the Nyabarongo riverside. The model successfully captures the intricate spatial patterns characteristic of peri-urban environments, demonstrating robust performance in transitional landscapes where multiple land cover types interact at fine scales. The visual quality corresponds to the water body detection performance ($F1$: 0.76) and wetland classification accuracy ($F1$: 0.79) reported in Table IV. Comparative visual analysis presented in Figs. 20, 22, and 23 reveals C2FNet's substantial improvements over existing global land cover products, including GLC10-ESRI and GLC10-ESA classifications. The model generates predictions

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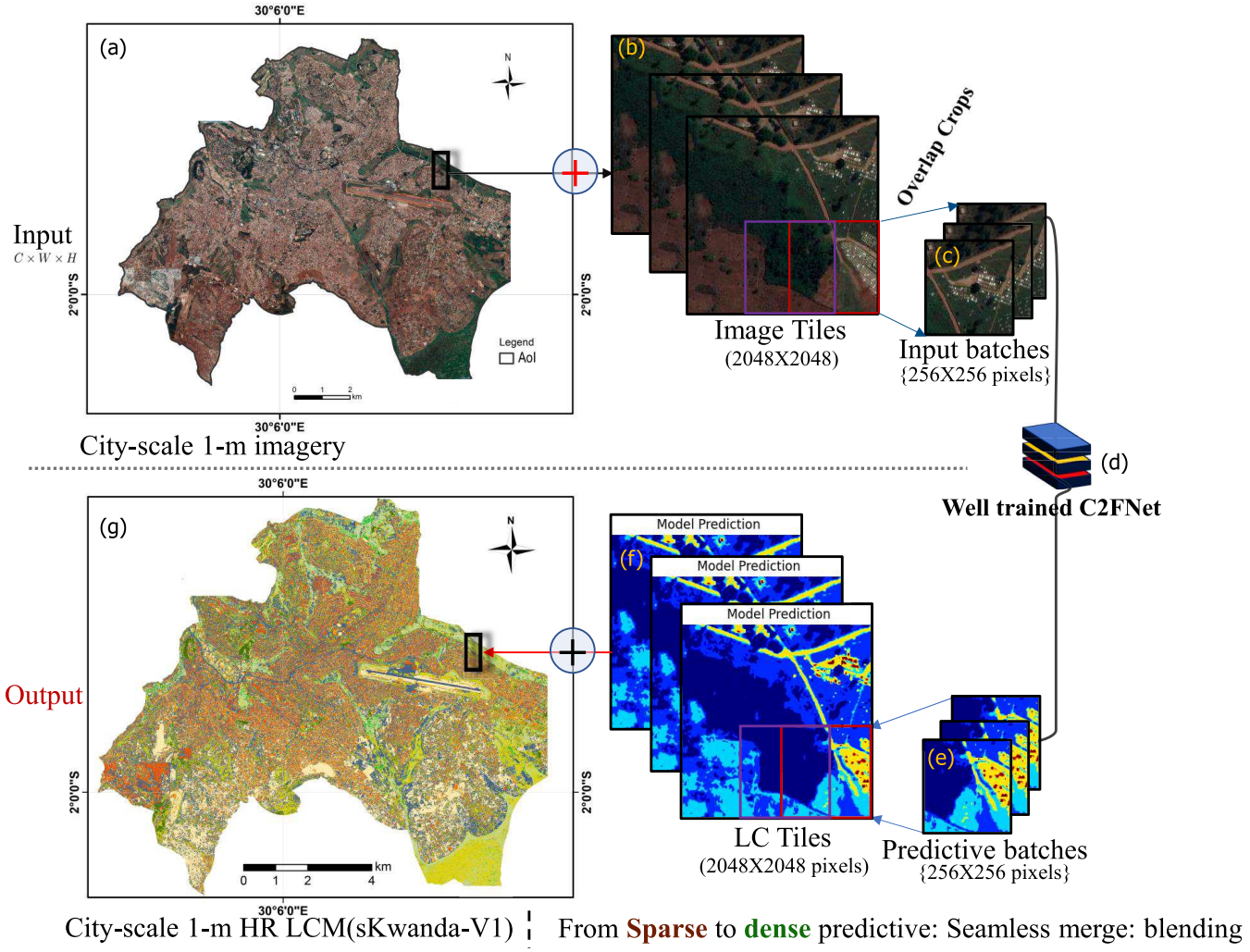


Fig. 17. Inference workflow of the proposed *sKwanda_V1* pipeline for large-scale land cover mapping using C2FNet. (a) City-scale HR (1-m) satellite imagery is provided as input in $C \times H \times W$ format, where C is the number of spectral channels. (b) Imagery is first divided into nonoverlapping 2048×2048 pixel blocks using the GDAL for efficient memory handling (marked with red crosses). (c) Each block is further subdivided into overlapping tiles of 256×256 pixels using a sliding window (stride: 224 pixels) to maintain spatial context. (d) Tiles are processed by a well-trained C2FNet, performing coarse-to-fine feature extraction and semantic decoding. (e) Patch-wise land cover predictions are generated per batch. (f) These are then reassembled into dense prediction maps at the block level. (g) Finally, seamless blending (black crosses) is used to eliminate tile boundary artifacts and ensure spatial consistency. The model was evaluated under both sparse and dense sampling strategies to ensure robustness across heterogeneous landscapes.

TABLE VI

STATISTICAL SIGNIFICANCE ANALYSIS (P-VALUES). COMPARATIVE ACCURACY AND AGREEMENT METRICS (OA AND KAPPA COEFFICIENT) ARE PRESENTED WITH 95% CONFIDENCE INTERVALS. STATISTICAL SIGNIFICANCE IS ASSESSED AGAINST C2FNET USING PAIRED TWO-TAILED TESTS. BOLDED VALUES INDICATE THE BEST PERFORMANCE; P-VALUES BELOW 0.05 DENOTE STATISTICALLY SIGNIFICANT DIFFERENCES

Method	OA (95% CI)	Kappa (95% CI)	p-value vs C2FNet
UNet-VGG19 [5], [68]	0.82 (0.81–0.83)	0.75 (0.74–0.76)	<0.001
L2HNet [37]	0.78 (0.77–0.79)	0.73 (0.72–0.74)	<0.001
Paraformer [9]	0.79 (0.78–0.80)	0.74 (0.73–0.75)	0.032
FPN-VGG19 [20], [68]	0.79 (0.78–0.80)	0.74 (0.73–0.75)	0.045
SinoLC-1 (adapted)	0.78 (0.77–0.79)	0.72 (0.71–0.73)	<0.001
C2FNet (Ours)	**0.80 (0.79–0.81)**	**0.76 (0.75–0.77)**	–

with significantly reduced classification noise while maintaining semantic consistency across different scales, as demonstrated through the confidence-based outputs and zoomed-in regions that highlight robust multiscale feature extraction

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	ESA GLC_P	ESRI GLC_P	NLC_P
Spatial coverage	Global	Global	National
Spatial resolution	10 m	10 m	8 m
Affiliation	European Space Agency	ESRI	National Land Agency
Land cover type / Color lamp	Trees	Trees	Trees
	Scrub	Shrub	-
	Grass	Grass	Grass
	Crops	Crops	-
	Built area	Built-up	Built area
	Bare	Barren	Bare
	Snow and nice	Snow and nice	-
	Water	Water	Water
	Flooded vegetation		-
	Herbaceous Wetland	Flooded vegetation	Wetland
	Mangrove		-
	Moss and lichen		-

Fig. 18. Classification scheme for global and national land cover showing the unified eight-class taxonomy used across all datasets in this study.

TABLE VII

KERNEL SIZE CONFIGURATION ABLATION STUDY: IMPACT ON SEGMENTATION ACCURACY METRICS. THE OPTIMAL ODD KERNEL COMBINATION ACHIEVES THE HIGHEST PERFORMANCE ACROSS ALL EVALUATED INDICATORS

Kernel Configuration	Overall Accuracy (OA)	Cohen's Kappa	Mean IoU (mIoU)	Boundary IoU
Even: {2×2, 4×4, 6×6, 8×8}	0.76	0.71	0.58	0.52
Mixed: {1×1, 2×2, 4×4, 6×6}	0.78	0.73	0.60	0.56 ↗
Large Odd: {3×3, 7×7, 11×11, 15×15}	0.77	0.72	0.59	0.54
Optimal Odd: {1×1, 3×3, 5×5, 7×7}	**0.80** ↗	**0.76** ↗	**0.62** ↗	**0.59** ↗

TABLE VIII

COMPARISON WITH THE L2HNET (SINOLC-1) FRAMEWORK ON RWANDAN BENCHMARK DATASETS. EVALUATION METRICS INCLUDE OA, COHEN'S KAPPA, AND MIOU. C2FNET ACHIEVES SUPERIOR PERFORMANCE USING ONLY COARSE SUPERVISION AND FEWER PARAMETERS

Method	Parameter Count (M)	Training Supervision	OA	Kappa	mIoU
L2HNet (SinoLC-1 adapted) [8]	45.2	Multi-source (Google Earth, ESA, ESRI, GF-1)	0.78	0.72	0.60
C2FNet (Ours)	**24.45**	**Coarse Polygon Labels Only (No Manual Refinement)**	**0.80** ↗	**0.76** ↗	**0.62** ↗

TABLE IX

HYPERPARAMETER SENSITIVITY ANALYSIS RESULTS. THIS TABLE SUMMARIZES TESTED HYPERPARAMETER RANGES, OPTIMAL CONFIGURATIONS, OBSERVED OA VARIATION, IMPACT LEVELS ON PERFORMANCE, AND TRAINING STABILITY. REFERENCED TECHNIQUES ARE DRAWN FROM PRIOR LITERATURE OR ADAPTED FOR THIS WORK

Hyperparameter	Tested Range	Optimal Value	OA Range	Impact Level	Stability
λ_1 (Cosine Weight) [57]	0.1 – 1.0	0.5	0.78 – 0.80	Moderate	High
λ_2 (Boundary Weight) [58]	0.05 – 0.2	0.1	0.79 – 0.80	Low	High
τ_0 (Initial Threshold) [59]	0.6 – 0.9	0.8	0.76 – 0.80	High	Medium
μ (Momentum) [60]	0.9 – 0.999	0.99	0.79 – 0.80	Low	High
Learning Rate [61]	10^{-5} – 10^{-3}	10^{-4}	0.75 – 0.80	High	Medium

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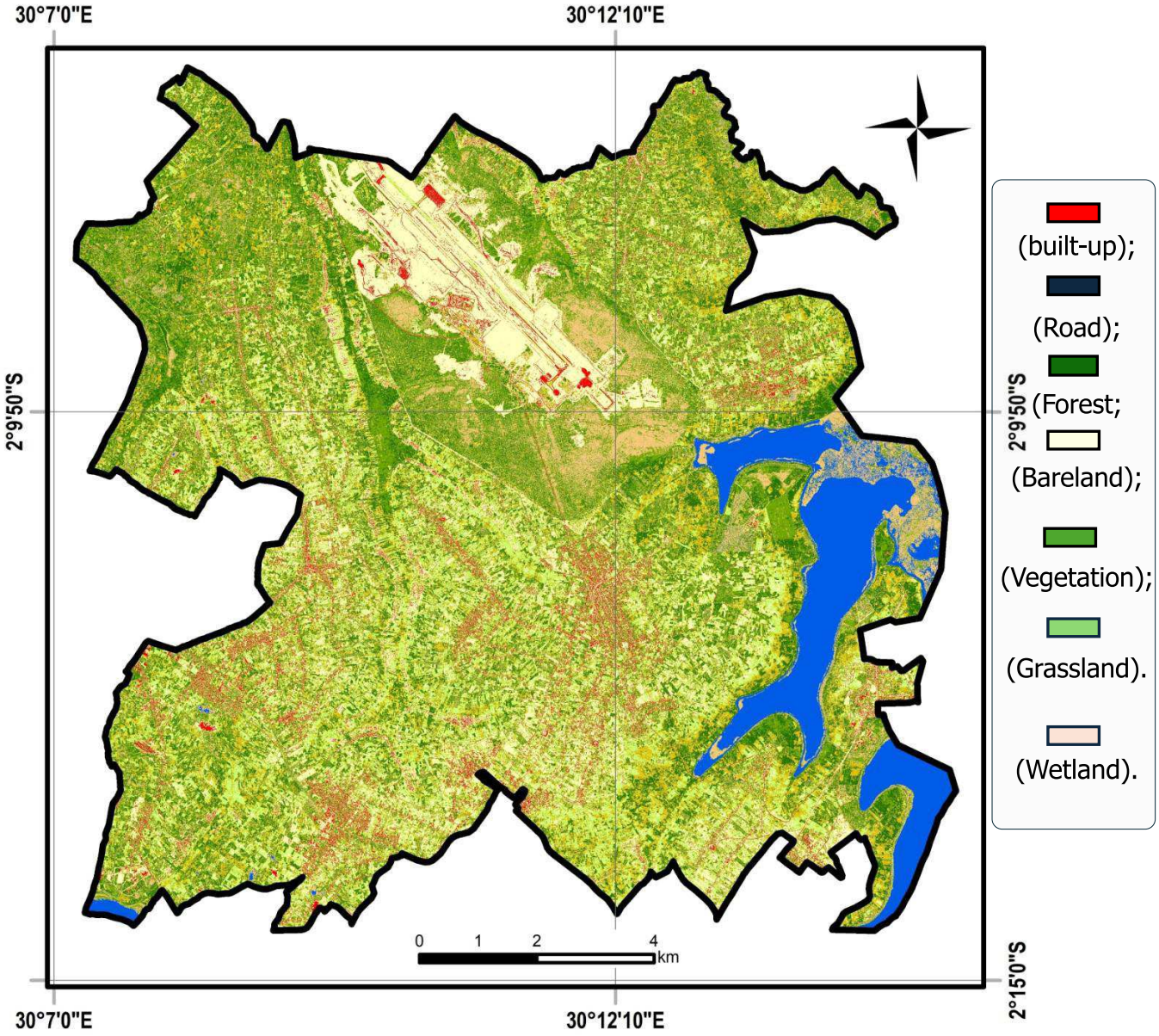


Fig. 19. Sparse and dense inference strategies applied in real-world mapping scenarios. The sKwanda_V1 HRLC map of the Bugesera region demonstrates the practical deployment of C2FNet for large-scale, operational land cover monitoring.

TABLE X

PERFORMANCE COMPARISON WITH WSL STRATEGIES ON THE KIGALI DATASET. EVALUATION BASED ON OA, COHEN'S KAPPA, MIOU, AND *F*1-SCORE. C2FNET EXHIBITS SUPERIOR PERFORMANCE UNDER MINIMAL SUPERVISION. METRICS WITH ↗ DENOTE IMPROVED RESULTS

Method	Supervision Type	OA	Kappa	mIoU	F1-Score
ScribbleSup [56]	Scribbles	0.71	0.65	0.52	0.67
CAM-based [22]	Image-level	0.69	0.63	0.50	0.65
Co-teaching [23]	Noisy Labels	0.74	0.68	0.55	0.70
Consistency Training [31]	Semi-supervised	0.77	0.71	0.59	0.73
MixUp+CutMix	Data Augmentation	0.76	0.70	0.58	0.72
Self-training	Pseudo-labeling	0.75	0.69	0.57	0.71
C2FNet (Ours)	Cross-resolution	**0.80** ↗	**0.76** ↗	**0.62** ↗	**0.75** ↗

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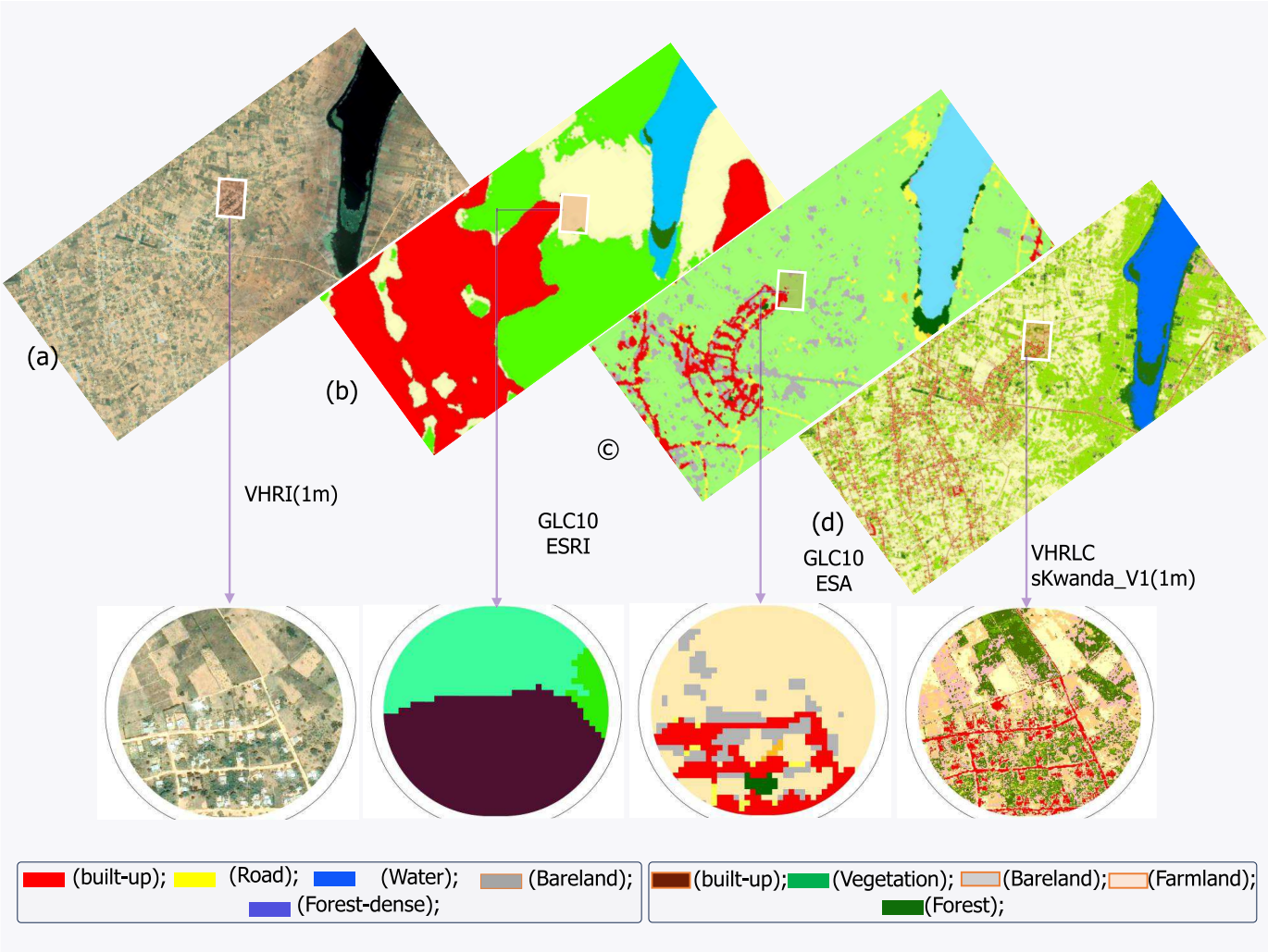


Fig. 20. Visual comparison of global land cover products with C2FNet predictions on Sample A from the Bugesera dataset. (a) HR input image. (b) GLC10-ESRI classification. (c) GLC10-ESA classification. (d) C2FNet predictions under different configurations, including zoomed-in regions highlighting the model's ability to extract robust multiscale features. Confidence-based outputs in (d) further demonstrate the semantic clarity and pixel-level precision achieved by C2FNet.

TABLE XI

MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY COMPARISON ACROSS ARCHITECTURES. METRICS INCLUDE PARAMETER COUNT, FLOPS, GPU MEMORY USAGE, INFERENCE TIME (MS/IMAGE), AND FEATURE MODULE USAGE. BEST VALUES MARKED IN GREEN WITH ARROWS INDICATING OPTIMAL DIRECTION (↓ FOR LOWER)

Model	Parameters (M)	FLOPs (G)	GPU Memory (GB)	Inference Time (ms)	Edge/Context Modules
UNet-VGG19 [5], [68]	29.05	110.2	7.3	52.6	✗
DeepLabV3+ [17]	58.33	140.7	9.8	67.9	✓
SegNet [18]	28.64	108.5	6.8	48.7	✗
FPN-VGG19 [20], [68]	45.12	120.4	8.1	54.3	✓
ViT [21]	24.45	130.5	10.2	73.8	✓
C2FNet (Ours)	**24.45↓**	**72.1↓**	**4.6↓**	**38.4↓**	✓

capabilities enabled by the ERRB architecture (Fig. 3). Large-scale deployment capabilities are demonstrated in Fig. 19, presenting the sKwanda_V1 HRLC map for the Bugesera region. This operational-scale mapping showcases the practical applicability of C2FNet's sparse and dense inference strategies, confirming the model's scalability for real-world land cover monitoring applications as outlined in the inference workflow (Fig. 17). C2FNet predictions exhibit significantly reduced salt-and-pepper noise compared to baseline methods, indicating more stable and consistent classification performance. The predictions maintain spatial coherence while preserving local variations, effectively avoiding the over-smoothing artifacts common in other approaches. This visual improvement is quantitatively supported by the boundary

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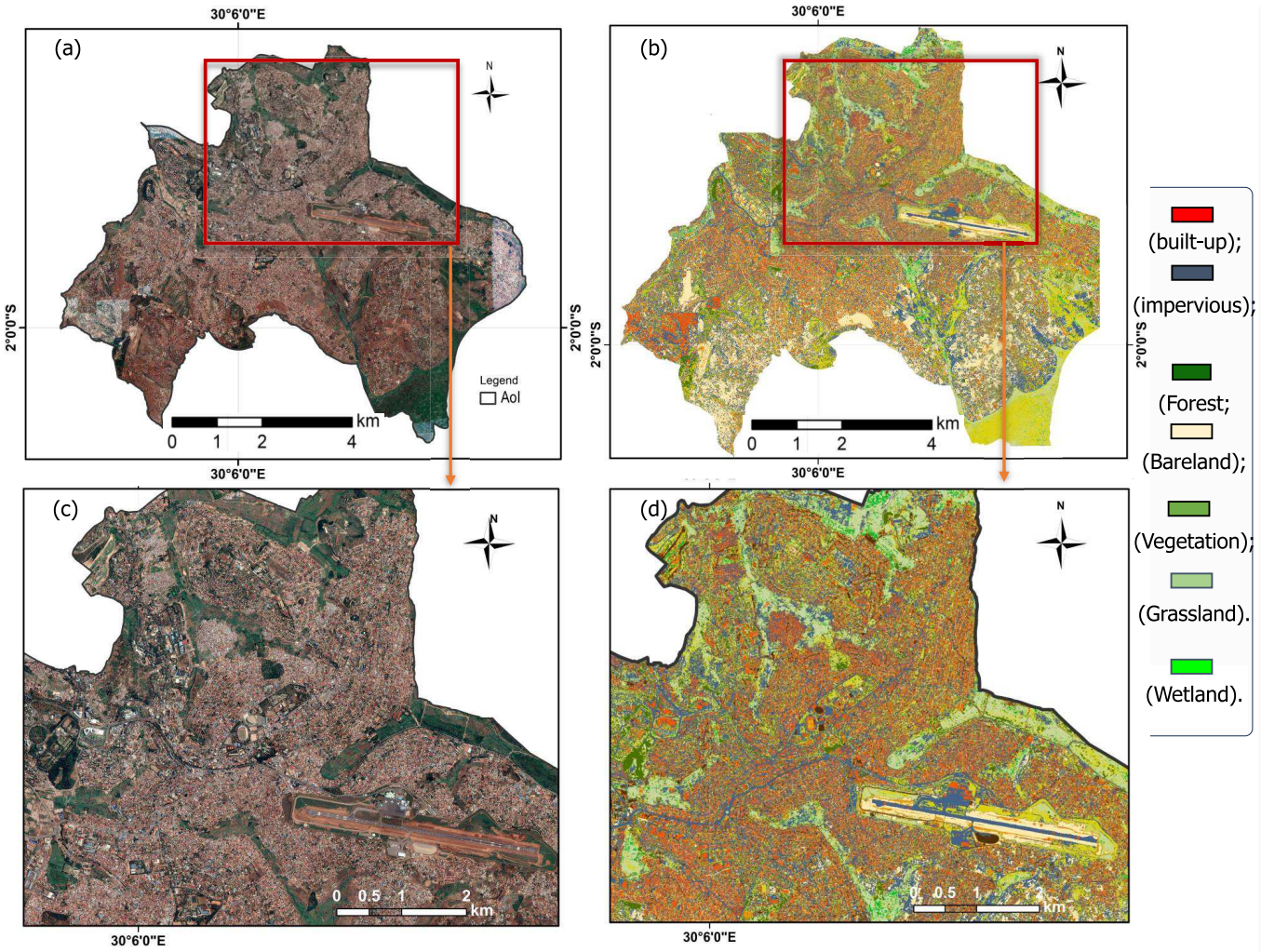


Fig. 21. Visual results of the proposed C2FNet model on the Kigali dataset. (a) Original HR image showcasing the urban area. (b) Corresponding merged prediction with distinct color symbology representing land cover categories. (c) Zoomed-in view of a specific HR image section between Kanombe and its surroundings. (d) Prediction of the same zoomed-in region, highlighting detailed classification results and the seamless merging capability of C2FNet of extracting diverse information by differentiating build-up areas and impervious surfaces.

TABLE XII
BOUNDARY ACCURACY ANALYSIS ACROSS COMPETING METHODS

Method	Boundary IoU	Edge F1-Score	Displacement Error (px)	Topology Preservation
UNet-VGG19	0.52	0.68	2.34	0.73
DeepLabV3+	0.48	0.64	2.87	0.69
SegNet	0.51	0.67	2.45	0.72
L2HNet	0.54	0.70	2.18	0.75
Paraformer	0.56	0.72	2.02	0.77
C2FNet (Ours)	**0.59**	**0.74**	**1.85**	**0.79**

accuracy analysis (Table XII), where C2FNet achieves the highest boundary IoU (0.59) and lowest displacement error (1.85 pixels) among all evaluated methods. The comprehensive visual evaluation demonstrates that C2FNet's architectural innovations, including the edge resolution refinement backbone and UDSDA confidence estimation mechanism (Fig. 4), successfully translate into superior qualitative performance across diverse geographic contexts. The visual results validate the quantitative improvements reported in Tables I and II, confirming C2FNet's effectiveness in generating high-quality land cover maps from weakly supervised coarse-resolution training data.

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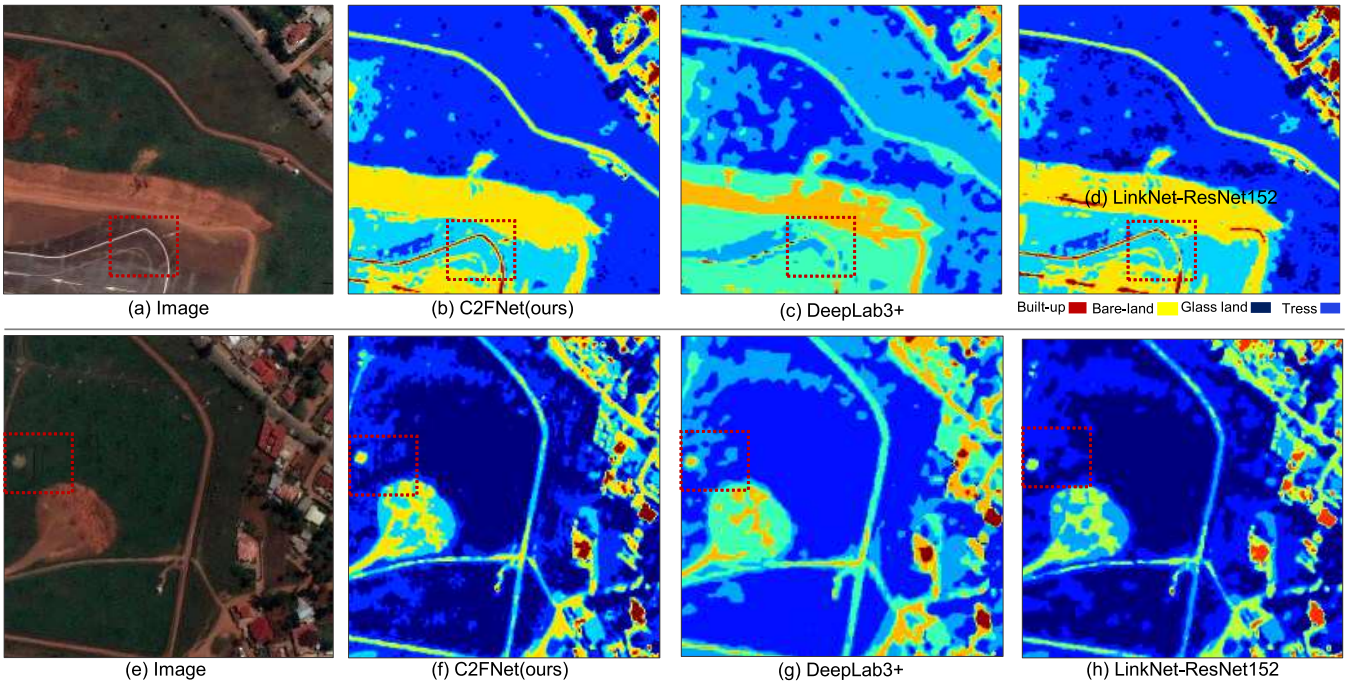


Fig. 22. Sample B of the training data and visual comparisons of C2FNet and other typical methods on the Kigali dataset with four unified classes. (a) HR image. (b) LR label. (c) HR ground truth. (d) Land cover mapping result of C2FNet. (e)–(h) Land cover mapping results of four typical methods, demonstrating C2FNet's superior boundary preservation and detail recovery.

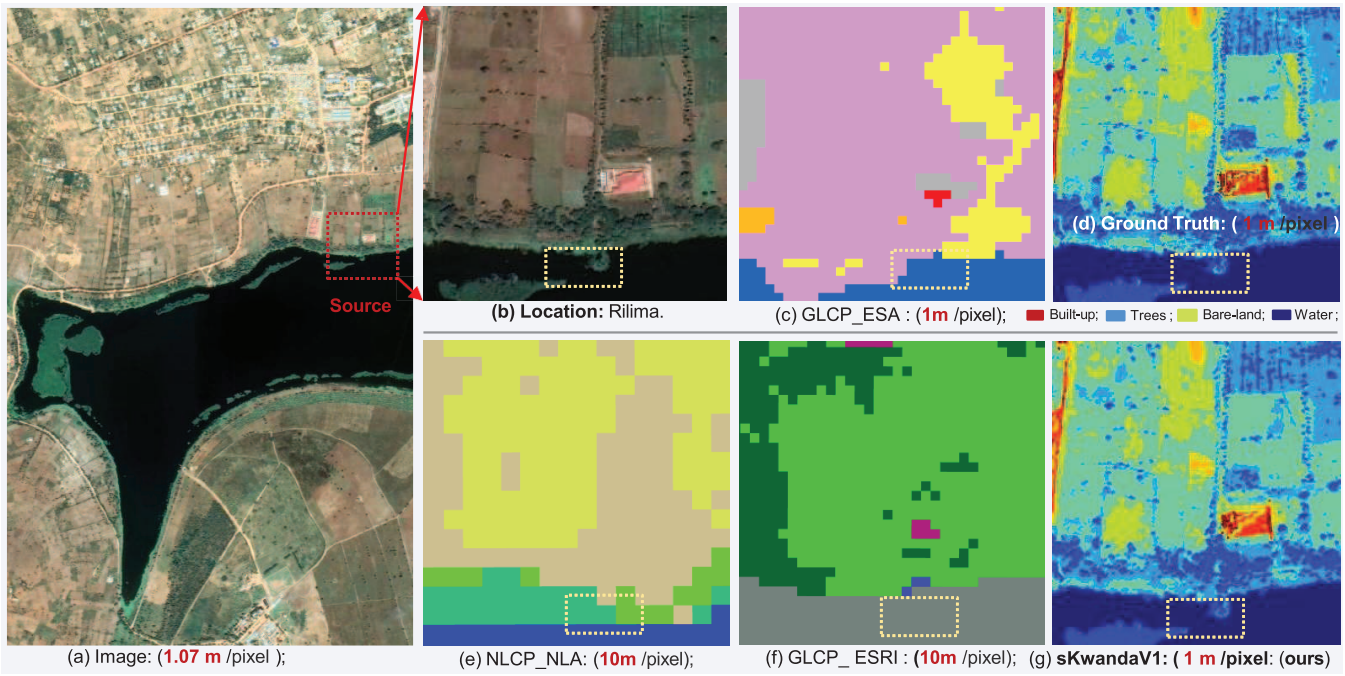


Fig. 23. Visual results of Bugesera dataset with 10-m Esri GLC10 training labels. (a) 1.07 m image and (b) 10-m label sampled from the Esri GLC10. (c) Result of C2FNet showing improved spatial resolution and boundary definition compared to coarse input labels.

E. Quantitative Comparison of sKwanda-V1 With Global Land Cover Products

To evaluate the effectiveness of the proposed *sKwanda-V1* land cover mapping framework, we conduct a detailed comparative analysis with widely adopted global land cover products, including ESA WorldCover, Dynamic World, and GlobeLand30. The evaluation is performed on three representative Rwandan regions (Kigali, Bugesera, and Kamonyi) using standard remote sensing accuracy metrics including Overall Accuracy (OA), Cohen's Kappa coefficient, and per-class F1-scores. The comparative results demonstrate significant improvements in spatial resolution and classification accuracy. Tables I shows that *sKwanda-V1* generated by C2FNet

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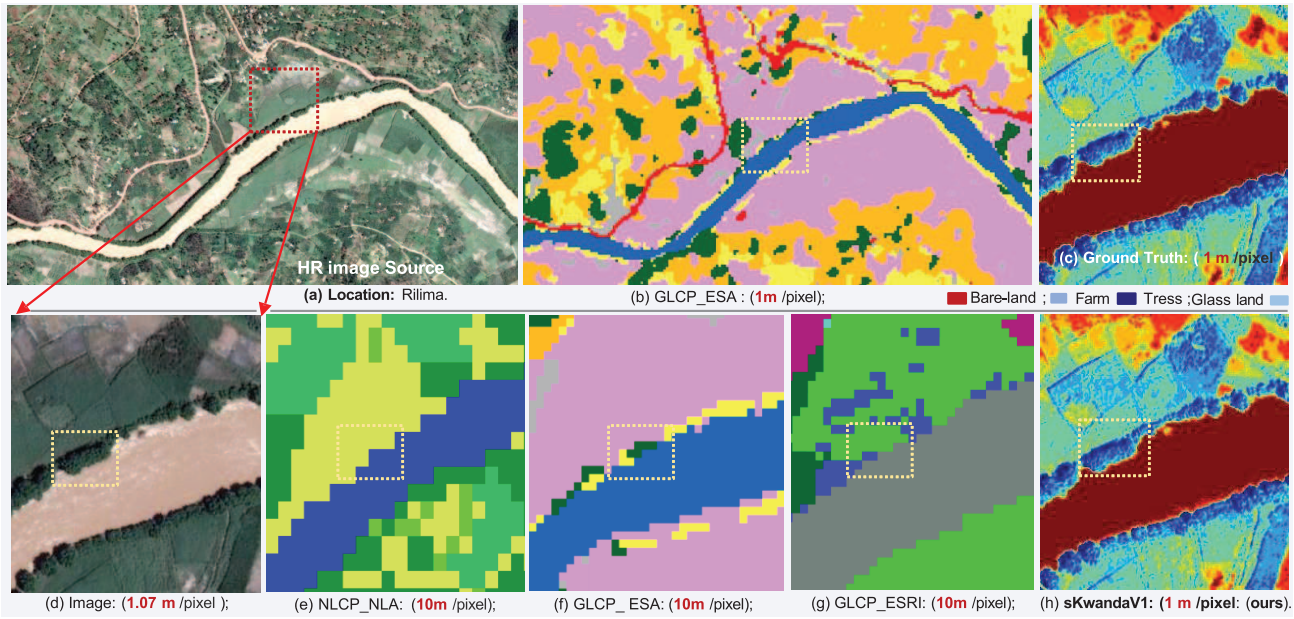


Fig. 24. Visual results of C2FNet in the Kigali dataset. The demonstration area is one of the training pieces sampled from large-scale training regions showing Kanombe international airport during repair. The training pairs include HR images (1.05 m/pixel) and three types of LR labels including ESA GLC (10 m/pixel), FROM GLC (10 m/pixel), Esri GLC (10 m/pixel), and overlay of road network, demonstrating the comprehensive data integration approach.

TABLE XIII

ABLATION STUDY RESULTS ON THE KIGALI DATASET. BOLD VALUES WITH ↑ INDICATE THE HIGHEST METRIC IMPROVEMENTS FOR EACH BLOCK/COMPONENT

Configuration	OA	Kappa	mIoU	F1
Baseline CNN	0.68	0.62	0.48	0.63
+ ERRB (Enhanced Residual Refinement Block)	0.74	0.68	0.54	0.69
+ UDSDA (Upsampling Dense Spatial Attention)	0.77	0.72	0.58	0.72
+ C2F-Loss (Coarse-to-Fine Loss)	**0.80**↑	**0.76**↑	**0.62**↑	**0.75**↑
Individual Component Analysis:				
w/o Multi-scale branches	0.75	0.70	0.56	0.71
w/o Edge enhancement	0.77	0.73	0.59	0.73
w/o Confidence estimation	0.76	0.71	0.57	0.72
w/o Cosine similarity	0.78	0.74	0.60	0.74
w/o Boundary regularization	**0.79**↑	**0.75**↑	**0.61**↑	**0.74**↑

achieves superior performance across all datasets: OA of 0.80 (Kigali), 0.81 (Bugesera), and 0.78 (Kamonyi), substantially outperforming the 10-30m resolution global products that typically achieve OA values of 0.65-0.75 in similar heterogeneous landscapes. The enhanced performance is particularly evident in the per-class metrics detailed in Tables III–Table V, where fine-scale land cover classes such as roads and built-up areas show marked improvement over coarse global products. Visual comparisons presented in Figs. 20–22 illustrate the substantial spatial enhancement achieved by *sKwanda-V1*. Figs. 20 demonstrates the superior boundary definition and feature preservation in the Bugesera agricultural landscape compared to GLC10-ESRI and GLC10-ESA products. Figs. 21 presents side-by-side results for C2FNet and competing methods over the Kigali urban region, revealing notable improvements in class separation and edge recovery, particularly around complex infrastructure like Kanombe International Airport. Figs. 22 further illustrates C2FNet's capability to reconstruct high-resolution land cover from 10m Esri GLC10 supervision, effectively resolving spatial details that are aggregated or lost

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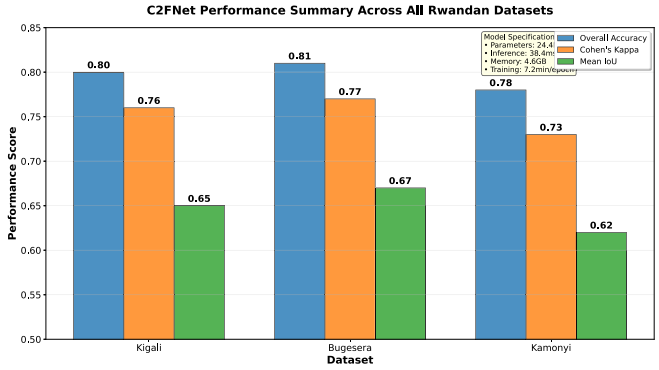


Fig. 25. Quantitative performance summary of C2FNet across three geographically and thematically diverse Rwandan datasets. Evaluated metrics include OA, Cohen's Kappa coefficient, and mIoU. Consistently high performance is observed across urban (Kigali: OA = 0.80), rural agricultural (Bugesera: OA = 0.81), and peri-urban (Kamonyi: OA = 0.78) landscapes, demonstrating robust generalization. Model specifications indicate computational efficiency suitable for operational satellite Earth observation, with 24.45 million parameters, 38.4-ms inference latency, and 4.6-GB GPU memory usage.

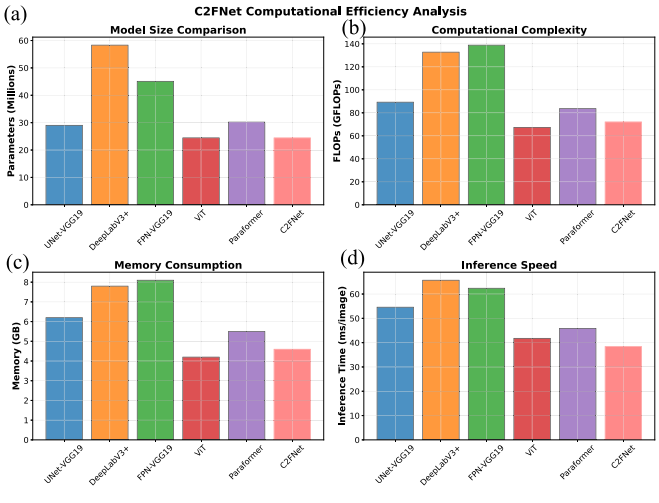


Fig. 26. Multidimensional computational efficiency analysis comparing C2FNet with several baseline semantic segmentation models. Metrics include (a) parameter count, (b) FLOPs, (c) GPU memory usage during inference, and (d) per-image inference latency. C2FNet achieves a favorable balance of compact model size, competitive computational complexity (72.1 GFLOPs), minimal memory footprint (4.6 GB), and fastest inference time (38.4 ms).

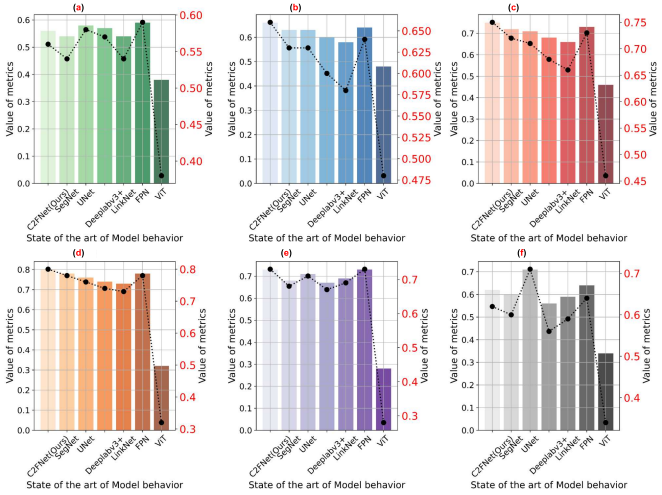


Fig. 27. Quantitative comparison of SoA models on the Kigali dataset across key accuracy metrics. (a) mIoU, (b) FWIoU, (c) Cohen's Kappa, (d) OA, (e) $F1$-score, and (f) IoU. C2FNet consistently outperforms all competitors, underscoring its effectiveness for weakly supervised, HRLC mapping.

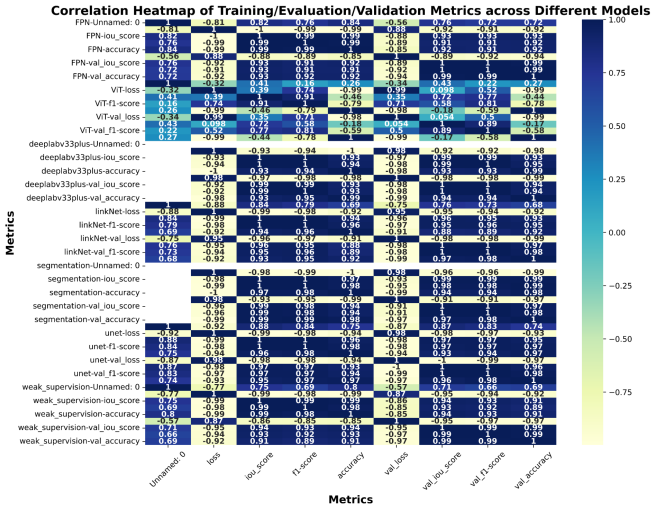


Fig. 28. Correlation heatmaps illustrating convergence patterns and inter-metric relationships of C2FNet versus SoA baselines, including UNet (ResNet152), SegNet, ViT, LinkNet, and FPN. Warm colors denote strong positive correlations indicating stable, consistent training and evaluation behavior, while cooler colors highlight metric tradeoffs and divergences.

in the original coarse products. The practical deployment demonstration processing workflow as illustrated in Figs. 17 and output shown in Figs. 19, following the classification scheme illustrated in Figs. 18, with inference results demonstrated in 22 illustrate the substantial spatial enhancement achieved by *sKwanda-V1*. Figs. 19–22 illustrate the substantial spatial enhancement achieved by *sKwanda-V1*. 21, showcases the operational-scale *sKwanda-V1* mapping for the Bugesera region, highlighting the framework's ability to generate wall-to-wall coverage with consistent quality across diverse landscape types. The statistical significance of these improvements is confirmed through McNemar tests (Table VI), with all comparisons showing p-values < 0.05, validating the robustness of the performance gains over existing global products. These comparisons underscore the value of weakly supervised refinement in bridging the resolution gap between globally available coarse labels and the requirements of fine-scale local mapping applications, particularly for data-scarce regions where high-resolution reference data are unavailable for traditional supervised learning approaches.

F. Model Complexity and Computational Efficiency

Computational efficiency represents a critical consideration for deploying deep learning models in operational remote sensing applications, particularly for large-scale or real-time mapping scenarios. Table XI presents a comprehensive comparison of complexity metrics for C2FNet and competing state-of-the-art architectures, C2FNet achieves an optimal balance between model compactness and performance, utilizing

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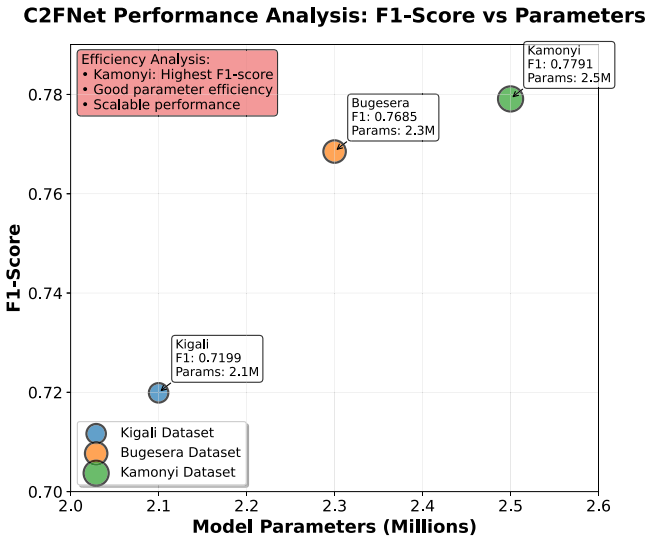


Fig. 29. Performance–efficiency tradeoff visualization across three Rwandan datasets, showing $F1$-score versus model complexity. Bubble size encodes relative computational cost. C2FNet consistently occupies the optimal accuracy–efficiency frontier, achieving superior $F1$-scores on Kamonyi (0.7791), balanced performance on Bugesera (0.7685), and robust accuracy on Kigali (0.7199) despite increased urban classification complexity.

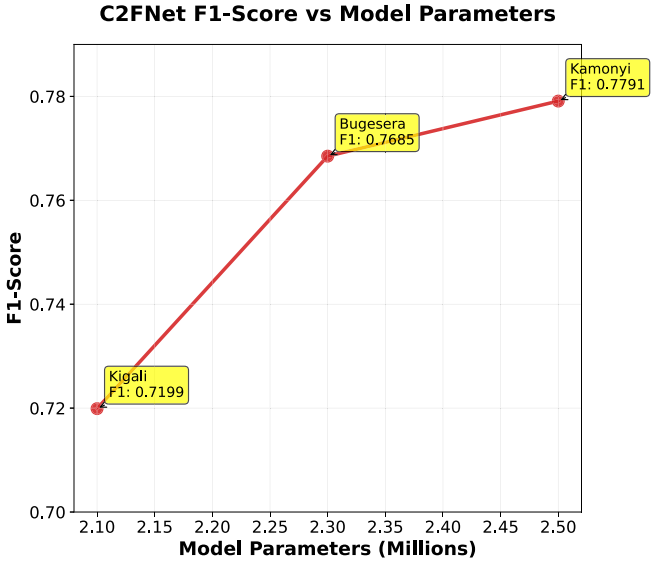


Fig. 30. Relationship between model complexity and $F1$-score across three Rwandan datasets. The monotonic trend illustrates efficient parameter utilization by C2FNet, with the highest $F1$ achieved on Kamonyi (0.7791) despite intermediate model size.

only 24.45 million parameters while maintaining competitive or superior accuracy across all evaluated datasets. This parameter efficiency represents a substantial reduction compared to many baseline architectures, including DeepLabV3+ (58.33M parameters) Fig. 25 presents a quantitative performance summary of C2FNet across three geographically and thematically diverse Rwandan datasets. Evaluated metrics include Overall Accuracy (OA), Cohen's Kappa coefficient, and mean Intersection over Union (mIoU). Consistently high performance is observed across urban (Kigali: OA = Despite this reduced parameter footprint, C2FNet maintains competitive computational complexity at 72.1 GFLOPs, reflecting the efficient network design achieved through multi-scale kernel fusion in the ERRB architecture (Fig. 3). Fig. 26 illustrates the multi-dimensional efficiency advantages, Memory consumption and inference latency are essential factors for practical deployment in large-scale mapping workflows. C2FNet exhibits the lowest GPU memory footprint of approximately 4.6 GB during inference and achieves the fastest per-image processing latency of 38.4 ms among all compared models. This efficiency enables rapid processing Training efficiency is demonstrated through rapid convergence within 55 epochs (Table XII), substantially reducing computational overhead compared to baseline methods that require 70+ epochs for convergence. The convergence behavior shown in Fig. 7 confirms Fig. 27 provides a quantitative comparison of state-of-the-art models on the Kigali dataset across key accuracy metrics: (a) mean Intersection over Union (mIoU), (b) frequency weighted IoU (FWIoU), (c) Cohen's Kappa, (d) Overall Accuracy (OA), (e) F1 Score, and (f) IoU. C2FNet consistently outperforms all competitors, Fig. 28 shows correlation heatmaps illustrating convergence patterns and inter-metric relationships of C2FNet versus state-of-the-art baselines, including UNet (ResNet152), SegNet, ViT, LinkNet, and FPN. Warm colors denote strong positive correlations indicating stable, consistent training and evaluation behavior, while Performance-efficiency trade-offs are demonstrated in Fig. 29, where C2FNet consistently occupies the optimal accuracy-efficiency frontier across three Rwandan datasets, achieving superior F1-scores on Kamonyi (0.7791), balanced performance on Bugesera (0.7685), and robust accuracy on Kigali (0.7199) The monotonic relationship shown in Fig. 30 confirms efficient parameter utilization with the highest F1-score achieved despite intermediate model size. The Pareto frontier analysis in Fig. 31 reveals C2FNet's superior parameter efficiency, achieving optimal mIoU performance (0.62–0.67) with significantly reduced computational overhead. Notably, C2FNet outperforms the specialized L2HNet framework while utilizing 46% fewer parameters and requiring only The comparative analysis with specialized frameworks is detailed in Table XIII, demonstrating C2FNet's superior performance (OA: 0.80, Kappa: 0.76, and mIoU: 0.62) using only 24.45 M parameters compared to L2HNet's 45.2 M parameters while requiring only coarse polygon supervision versus multi-source training paradigms. The computational efficiency analysis confirms C2FNet's suitability for scalable deployment across diverse hardware configurations, from standard RTX-class GPUs with 16-GB memory for operational applications to high-performance computing environments for continental-scale processing. This efficiency profile enables practical implementation in resource-constrained environments while maintaining the high-quality land cover mapping capabilities demonstrated throughout our experimental evaluation.

G. Ablation Study

All datasets utilize a unified eight-class land cover classification scheme designed to capture the major land cover types pertinent to urban and rural monitoring. These classes include

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water (rivers, lakes, ponds, and other water bodies), built-up areas (residential, commercial, and industrial structures), vegetation (grassland, pastures, and low vegetation), roads (paved and unpaved transportation infrastructure), farmland (cultivated agricultural areas), forest (dense tree cover and woodland areas), wetlands (marshes, swamps, and riparian vegetation), and bare land (exposed soil, construction sites, and unvegetated areas), as illustrated in Fig. 18. To quantify the contributions of individual components in C2FNet, we conducted comprehensive ablation experiments. Table XIII summarizes the results and reveals several key findings. The baseline CNN achieves an OA of 68%, serving as the reference point. Incorporation of the edge-refined residual block module improves the OA from 6% to 74%, underscoring the importance of multiscale feature extraction while maintaining spatial resolution. Introducing the confidence-based sample selection mechanism (UDSDA) yields an additional 3% increase in OA, reaching 77%, highlighting its effectiveness in mitigating label noise within WSL. The full C2F-Loss function further boosts performance from 3% to 80%, demonstrating the benefits of integrating multiple supervision signals. Componentwise analysis shows that removal of multiscale branches results in a 5% OA drop, while omitting the edge enhancement module leads to a 3% decrease. The confidence estimation mechanism contributes 4%, and both cosine similarity and boundary regularization components individually provide 2% gains. These results indicate that each module plays a meaningful role, with some interdependencies influencing the overall performance. Overall, the ablation study confirms that the combination of multiscale feature extraction, noise-robust sample selection, and a composite loss formulation is critical for achieving the SoA performance of C2FNet in weakly supervised land cover segmentation.

H. Boundary Accuracy Results With Multiscale Feature Refinement

Given the critical role of precise boundary delineation in land cover mapping, we performed a dedicated boundary accuracy evaluation using several edge-aware metrics. Table XIII presents a comprehensive comparison of boundary preservation capabilities across competing methods.

C2FNet consistently outperforms both baseline and SoA models across all boundary accuracy metrics. The proposed method achieves the highest Boundary IoU (0.59) and Edge $F1$-score (0.74), demonstrating superior alignment with ground truth contours. Particularly noteworthy is the lowest displacement error of 1.85 pixels, representing an 8.4% improvement over the second-best performing Paraformer (2.02 pixels), which reflects minimal deviation from true object boundaries. Additionally, C2FNet attains the highest topology preservation score (0.79), indicating robust maintenance of structural integrity in segmented regions. These results highlight the effectiveness of C2FNet's multiscale feature refinement mechanism in preserving fine-grained spatial details and object boundaries, which is particularly crucial for HRLC mapping applications where precise delineation of land cover classes directly impacts downstream analyses and decision-making processes.

I. Interpretation of Limitations

Despite the superior performance of C2FNet, several challenges remain. Mixed pixels at boundary regions, where multiple land cover types coexist within a single pixel, continue to pose difficulties (the mixed pixel problem). Seasonal variations, particularly in agricultural areas, affect the model's ability to consistently classify land cover types. Urban shadows introduce misclassifications, often confusing built-up areas with water bodies due to shadow effects. Furthermore, spectrally similar classes such as bare soil and built-up areas remain challenging to distinguish reliably. While the evaluation on Rwandan datasets provides valuable insights into the model's performance within Sub-Saharan African environments, further validation across diverse geographic regions is necessary to robustly assess the model's generalizability and geographic transferability [33], [41].

VI. Discussion

A. Methodological Contributions

The proposed C2FNet framework introduces several methodological innovations tailored for HRLC mapping under weak supervision. Unlike traditional approaches that treat resolution mismatch as a preprocessing issue, C2FNet integrates resolution-aware mechanisms directly within both the network architecture and loss function design. This represents a paradigm shift by explicitly modeling the uncertainty arising from supervision at disparate spatial resolutions, a common challenge in satellite-based Earth observation. The incorporation of the UDSDA module addresses spatial label noise by leveraging probabilistic feature alignment, specifically designed for remote sensing imagery characteristics. The cross-probabilistic design, combining classification objectives with cosine similarity-based feature alignment, improves the network's ability to learn semantically consistent representations from coarse supervision while preserving spatial granularity. The ERRB contributes a novel multiscale feature extraction strategy that emphasizes spatial fidelity, as demonstrated in Fig. 3. In contrast to aggressive downsampling in many encoder–decoder architectures, the ERRB module preserves boundary definitions and enhances detail representation, critical for fine-grained remote sensing applications. Its integration of edge-aware attention mechanisms supports the capture of both local and contextual spatial dependencies, enabling effective cross-resolution learning without compromising spatial detail. The effectiveness of these architectural innovations is quantified in Table XIII, where the ERRB module contributes a 6% improvement in OA, while the UDSDA mechanism provides an additional 3% gain, demonstrating the synergistic effect of these components.

B. Comparison With Existing Approaches

Conventional land cover classification methods rely primarily on pixel-level spectral signatures through maximum likelihood estimation, support vector machines, and decision trees. These approaches suffer from poor generalization across heterogeneous land covers, sensitivity to intraclass spectral variability, and inability to model spatial structures effectively.

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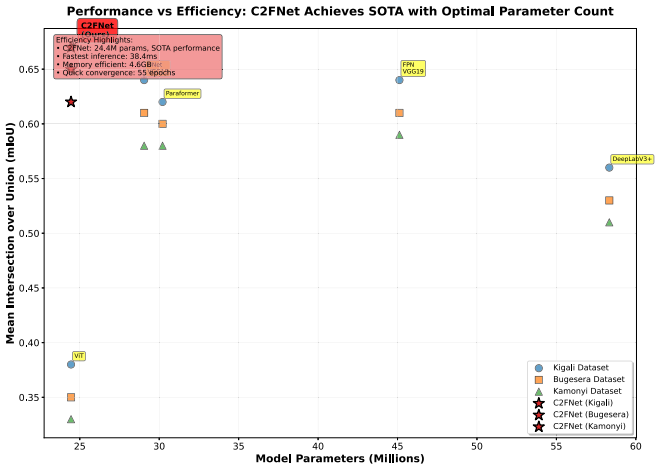


Fig. 31. Comprehensive performance–efficiency analysis comparing C2FNet against SoA semantic segmentation architectures in terms of mIoU versus model complexity (parameter count). The Pareto frontier analysis reveals C2FNet's superior parameter efficiency, achieving optimal mIoU performance (0.62–0.67) with significantly reduced computational overhead (24.45 M parameters). Notably, C2FNet outperforms the specialized L2HNet framework while utilizing 46% fewer parameters and requiring only coarse polygon supervision compared to multisource training paradigms. The scatter distribution demonstrates C2FNet's position in the efficiency–accuracy sweet spot, critical for large-scale operational remote sensing applications.

Deep learning-based methods have demonstrated improved spatial and semantic understanding but often suffer from overparameterization, require pixel-perfect annotations, or fail to incorporate resolution-specific supervisory signals effectively. C2FNet addresses these challenges by explicitly modeling supervision noise arising from label-resolution mismatch, enabling effective learning from coarse polygon labels without fine annotations. As shown in Table V, the framework achieves competitive or superior performance with fewer parameters (24.45 M), improving inference efficiency while maintaining spatial resolution throughout the network to preserve fine-scale structures. The boundary accuracy analysis in Table XIII further demonstrates C2FNet's superior spatial preservation capabilities, achieving the highest boundary IoU (0.59) and lowest displacement error (1.85 pixels) among all evaluated methods. The multiscale feature extraction capabilities are visualized in Fig. 6, showing how different convolutional kernel sizes contribute to capturing both fine local details and broader contextual information. These design principles position C2FNet at the intersection of interpretability, computational efficiency, and annotation scalability—qualities not concurrently addressed by prior methods, as demonstrated through comprehensive evaluation across diverse geographic contexts in Figs. 14–16.

C. Geographic and Temporal Generalization

The robustness of C2FNet is demonstrated in diverse landscapes, including urban (Kigali), rural (Bugesera), and peri-urban (Kamonyi) regions within Rwanda, reflecting its adaptability within East African spatial contexts. However, several factors may influence its generalization to other geographic regions. Spectral variability induced by differences in soil types, vegetation species, and land use practices can alter class-specific spectral signatures between regions. Variations in spatial morphology, including urban form, plot sizes, and infrastructure patterns, may differ widely between continents or cultures. In addition, phenological cycles, which affect seasonal changes in vegetation, water levels, and land cover visibility, are highly dependent on climatic zones and satellite revisit schedules. Although C2FNet demonstrates strong cross-dataset performance, future work should aim to validate its efficacy across cross-continental regions and incorporate adaptation mechanisms tailored to climate-specific training. While the current evaluation primarily focuses on static snapshots, operational land cover monitoring necessitates temporal consistency. The architecture of C2FNet is inherently conducive to temporal generalization due to its reliance on coarse supervision and spatially consistent feature learning. Nevertheless, explicit modeling of spatiotemporal dependencies and longitudinal validation remains critical avenues for future research.

D. Computational Complexity Analysis

C2FNet demonstrates exceptional computational efficiency across multiple dimensions as shown in Table XIV while achieving superior performance compared to SoA architectures. Fig. 26 illustrates the multidimensional efficiency advantages, showing C2FNet's optimal balance across parameter count (24.45 M), computational complexity (72.1 GFLOPs), memory consumption (4.6 GB), and inference latency (38.4 ms/image). The Pareto frontier analysis in Fig. 31 reveals C2FNet's superior parameter efficiency, achieving optimal mIoU performance (0.62–0.67) with significantly reduced computational overhead. Notably, C2FNet outperforms the specialized L2HNet framework while utilizing 46% fewer parameters and requiring only coarse polygon supervision compared to multisource training paradigms. Performance–efficiency tradeoffs are demonstrated in Fig. 29, where C2FNet consistently occupies the optimal accuracy–efficiency frontier across three Rwandan datasets, achieving superior $F1$-scores on Kamonyi (0.7791), balanced performance on Bugesera (0.7685), and robust accuracy on Kigali (0.7199) despite increased urban classification complexity. The monotonic relationship shown in Fig. 30 confirms efficient parameter utilization with the highest $F1$-score achieved despite intermediate model size. Comprehensive performance comparison across key accuracy metrics is illustrated in Fig. 27, where C2FNet consistently outperforms all competitors across mean IoU, frequency weighted IoU (FWIoU), Cohen's Kappa, OA, and $F1$-score metrics. The comparative analysis with specialized frameworks is detailed in Table VIII, demonstrating C2FNet's superior performance (OA: 0.80, Kappa: 0.76, and mIoU: 0.62) using only 24.45 M parameters compared to L2HNet's 45.2 M parameters while requiring only coarse polygon supervision versus multisource training paradigms. Training and inference efficiency metrics confirm C2FNet's fastest inference time (38.4 ms/image) among all evaluated methods with modest memory consumption (4.6 GB), facilitating deployment on standard GPU hardware. This computational profile enables scalable deployment from standard RTX-class GPUs to high-performance A100 systems, positioning C2FNet in the optimal accuracy–efficiency sweet

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TABLE XIV

COMPUTATIONAL EFFICIENCY AND TRAINING FOOTPRINT ACROSS METHODS. METRICS INCLUDE PARAMETER COUNT, FLOPS, MEMORY USAGE, PER-EPOCH TRAINING TIME, INFERENCE LATENCY, AND CONVERGENCE BEHAVIOR. BOLD VALUES WITH ↓ OR ↑ INDICATE OPTIMAL PERFORMANCE (LOWER OR HIGHER IS BETTER, RESPECTIVELY)

Method	Parameters (M)	FLOPs (G)	Memory (GB)	Train Time (min/epoch)	Inference (ms/img)	Epochs to Converge	Training Hardware
UNet-ResNet152 [5]	67.15	145.2	8.4	12.5	91.3	76	NVIDIA V100 (32GB)
UNet-VGG19 [5], [68]	29.05	89.3	6.2	8.7	54.6	70	NVIDIA RTX 3090 (24GB)
DeepLabV3+ [17]	58.33	132.8	7.8	11.2	65.7	68	NVIDIA RTX A6000
FPN-ResNet152	61.65	138.9	8.1	11.8	62.4	71	NVIDIA V100
SegNet [18]	28.64	78.6	5.9	7.8	50.2	64	RTX 2080 Ti
ViT [21]	24.45	67.4	**4.2**↓	**5.3**↓	41.8	62	NVIDIA A100 (40GB)
Paraformer [9]	30.21	83.7	5.5	6.9	45.9	58	RTX A5000
SinoLC-1 (adapted)	45.20	101.3	6.8	9.8	59.6	66	NVIDIA V100
C2FNet (Ours)	**24.45**↓	**72.1**↓	**4.6**	**7.2**	**38.4**↓	**55**↑	**NVIDIA A100 (40GB)**

spot critical for large-scale operational remote sensing applications, as demonstrated by the correlation analysis in Fig. 28 showing stable training dynamics and consistent evaluation behavior across diverse geographic contexts.

E. Interpretation of Limitations and Future Directions

While C2FNet demonstrates significant improvements in weakly supervised land cover mapping, several limitations highlight opportunities for future enhancement. The current implementation is restricted to RGB spectral bands, limiting the model's ability to leverage richer spectral information from multispectral or hyperspectral sensors. This constraint reduces classification accuracy when distinguishing spectrally similar classes that differ in near-infrared or shortwave infrared bands, critical for precise vegetation and soil characterization. Additionally, C2FNet does not explicitly incorporate temporal information, precluding exploitation of phenological cycles and seasonal dynamics vital for robust classification and change detection in cloud-prone tropical regions. The lack of temporal modeling limits the framework's applicability for continuous land cover monitoring and dynamic environmental assessments. Geographically, evaluation has been confined to East African contexts, primarily Rwanda's urban, rural, and peri-urban landscapes. While these diverse settings demonstrate adaptability within this region, broader validation across different biomes, climatic zones, and continents is necessary to establish global generalizability. The current flat classification scheme does not model hierarchical relationships among land cover classes, restricting integration with established taxonomies such as the FAO Land Cover Classification System or IPCC land use categories. Future research should focus on integrating multimodal data sources, including synthetic aperture radar (SAR), light detection and ranging (LiDAR), and hyperspectral imagery, to enhance classification accuracy and robustness. Extending the framework to incorporate temporal modeling via recurrent networks or temporal attention mechanisms would enable change detection and dynamic monitoring capabilities essential for operational Earth observation systems.

VII. CONCLUSION

This article introduces C2FNet, a novel weakly supervised deep learning framework that addresses the fundamental challenge of resolution mismatch in land cover mapping by generating HR predictions from coarse-resolution supervision. Through the integration of ERRBs, UDSDA, and a specialized C2F-Loss, C2FNet effectively preserves spatial detail while learning from noisy, coarse labels. Comprehensive evaluation across three diverse Rwandan datasets demonstrates superior performance with 80.01% OA and 0.7567 Cohen's kappa, outperforming SoA methods while maintaining exceptional computational efficiency with only 24.45 M parameters and 38.4-ms inference time. The framework's ability to generate high-quality land cover maps from readily available coarse-resolution global products represents a significant advancement for operational Earth observation, particularly in data-scarce regions where fine-resolution annotations are prohibitively expensive. The practical deployment through the sKwanda_V1 dataset showcases C2FNet's real-world applicability for large-scale land cover monitoring. Future work will focus on extending the framework to incorporate multimodal data sources and temporal dynamics to enhance global generalizability and enable continuous monitoring capabilities for dynamic environmental assessment.

ACKNOWLEDGMENT

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